

Supplementary Information for Two Modes, Not Many Types: The Structure of Football Player Roles in Season-Aggregate Data

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This document supports the article of the same name. Section and figure numbers prefixed with S refer to this document; unprefixed numbers refer to the article. Every number here is generated by the code in the accompanying repository and read into the document at build time, on the same terms as the article.

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S1 Extended methods

This section states in full the procedures the article summarises. Nothing here is a different analysis from the one reported there; it is the detail a reader would need in order to reproduce it without reading the code.

S1.1 The three clusterless generators

Let $X \in \mathbb{R}^{n \times d}$ be the standardised feature matrix for a scope, with n player-seasons and $d = 33$ features. Let $s_k(X)$ be the mean silhouette of the partition k -means returns at k clusters, computed on a fixed subsample of at most four thousand rows when n exceeds that, since the silhouette is quadratic in n .

The Gaussian generator draws n rows from $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ with the empirical mean and covariance of X . It matches the data in size, dimension and covariance and in nothing else: every marginal is symmetric and unbounded where the real per-90 rates are skewed and bounded below, and skew alone lifts the silhouette a partition can reach against this reference, which is why the copula generator stands beside it.

The copula generator transforms each column of X to normal scores by rank, estimates the correlation $\hat{R}$ of those scores, draws n rows from $\mathcal{N}(0, \hat{R})$, and maps each simulated column back through the empirical quantile function of the corresponding observed column. Every marginal distribution of X is therefore reproduced exactly, including skew, bounds, ties and any bimodality a single feature carries on its own, and the pairwise rank correlation is preserved up to sampling noise. What it destroys is joint structure beyond pairwise rank correlation, which is where two groups differing in several features at once would live. It tests whether the joint structure carries anything beyond the marginals, not whether the marginals are unimodal.

The uniform generator rotates X to its principal axes, draws n rows uniformly within the resulting bounding box, and rotates back. This is the reference distribution of Tibshirani et al. [1]. It assumes no distributional form at all. Every principal axis is filled evenly over its full observed range, so a bounded or heavy-tailed axis ranges over several times its standard deviation and the minor axes carry more spread than in the data, making the reference more isotropic than the sample it is matched to. That lowers its silhouette at two clusters, so for the silhouette it is a soft reference; its leading component is flat rather than peaked, which gives it the largest dip of the three, so for the dip statistic it is the hard one. Which generator is hardest in each scope is recorded in the metrics file as a result rather than assumed.

S1.2 Stability, replication and agreement with position

We draw two hundred bootstrap resamples, refit the clustering on each, and compute the adjusted Rand index between each bootstrap partition and the reference partition on the players common to both [2], accumulating a consensus matrix that records, for every pair of players, the fraction of resamples in which the two were assigned together. Clusters whose members co-assign below 0.6 on average are reported as unstable and described as such rather than being renamed or merged to make the account tidier.

Replication is tested separately and more stringently, because a resample reuses the same players. Each season and each league is standardised inside itself and clustered from scratch, borrowing no centre, no scale and no starting point from the pooled fit. Cluster labels are arbitrary per fit, so labels are aligned to the pooled partition on the contingency table before any label-dependent quantity is computed, and an assertion fails the run if an aligned agreement drops to chance. Three comparisons are reported because they answer different questions: an independent fit scored against the pooled

partition on the same rows; one season’s labels scored against another’s on the players present in both, which mixes replication with real change in those players; and one scope’s centroids imported into another scope’s players and scored against that scope’s own fit, which isolates replication. Each scope’s centroid-difference vector is also correlated with the pooled one, to check that the scopes recover the same axis rather than merely a similarly sized partition.

Agreement with the listed positions is quantified by normalised mutual information and by the adjusted Rand index, against both the position group and the full position string. Players whose cluster assignment disagrees with their listed group, and players whose Gaussian mixture posterior is split across components, are extracted as candidate hybrids.

S1.3 Feature-family ablation

A two-mode result on 33 features could in principle be carried by one block of them. The features are grouped into six families, shooting, chance creation, ball progression, pitch territory, defending and passing accuracy, with the grouping stated in the analysis code so that it can be audited and asserted to cover every feature exactly once. Each family is removed in turn and the pre-declared selection rule and the Gaussian null calibration are rerun on what remains, in every scope. The converse is run as well, each family alone, which shows which families carry the two-mode structure by themselves. For every ablation the paper records the number of clusters the rule selects, the standard score of the observed silhouette against the null, and the adjusted Rand index between the ablated partition and the full one on the same players, so that a surviving k of two can be distinguished from the same two modes surviving.

S1.4 Recovering and predicting results

The archive’s team tables carry goals and expected goals but not results, so wins, draws and losses are recovered by summing the goalkeeper table, which credits every match to the goalkeeper who played it. The recovery is audited: each league-season is checked for the right number of clubs, every club’s credited matches are checked against the schedule with the exceptions listed, goals for are taken from goals scored while each goalkeeper was on the pitch and accepted only where the same rows reproduce the goalkeeper table’s goals against to within one, and two published season records are asserted. Points per match, goal difference per match and league position follow.

Each team-season is then summarised by its minutes-weighted composition on the two-mode partition and on the six archetypes, using the eligible players, and by the minutes-weighted mean position of its players along the axis between the two centroids. Prediction is out of sample under three holdouts, one season at a time, one league at a time and one club at a time, the last being the strict one since no held-out team-season then shares a club with the training data, by ridge regression with the penalty chosen by cross-validation inside the training folds, comparing possession alone, the composition alone and both. The goalkeeper partitions are tested the same way, against the prediction that a partition sorting keepers by their teams will track points and a partition describing goalkeeping will not.

S1.5 Archetype profiling and outliers

For every retained cluster we compute the centroid profile as a standardised deviation from the mean of its position group, take the eight features with the largest absolute deviation as the distinguishing features, and list the ten players closest to the centroid and three players near the boundary. Archetype names are assigned only after inspecting these profiles and the players they contain, and each name is accompanied by a justification that cites the distinguishing features and the example

players. The names are hypotheses about what the statistical profile represents, not observations, and they carry no authority beyond the profile that produced them. Outliers are identified within position group with an isolation forest [3], reported alongside a Mahalanobis distance so that a distance-based and a partition-based notion of unusualness can be compared.

S1.6 Supervised validation

Logistic regression and gradient boosting, the latter with LightGBM [4], are trained to predict the position group from the feature set under five-fold stratified cross validation, and we report accuracy, macro F1 and the confusion matrix, every number computed from out-of-fold predictions. A majority-class baseline runs through the identical folds, so the reported lift is over the trivial model rather than over zero. SHAP values [5] on the boosted model identify which statistics distinguish each position; the model is fitted on every row and the attributions are computed on a stratified sub-sample, which is stated with the result. We also record, for each feature and class, the rank correlation between the feature’s value and the attribution it makes, which separates a class identified by what its members do from one identified by what they do not. Players the classifier places in the wrong group are cross-referenced against the candidate hybrids from the clustering, since agreement between two methods that fail in the same place is more informative than either alone.

A secondary ridge regression predicts non-penalty expected goals per ninety from the non-shooting features only, so that the fitted value represents the shot volume a player’s role would ordinarily generate and the residual represents the part not explained by role. We report the cross validated coefficient of determination and treat the residual ranking with heavy caution, because a residual from a model with modest explanatory power is mostly noise.

S1.7 Dimensionality reduction and the remaining analyses

Principal component analysis is fitted to the standardised feature set, both globally and separately within each position group. Uniform manifold approximation and projection [6] is fitted over a grid of twelve configurations, with the number of neighbours in $\{10, 15, 30, 50\}$ crossed with a minimum distance in $\{0.0, 0.1, 0.5\}$, and configurations are ranked by trustworthiness, which measures the degree to which points close in the embedding were also close in the original space. Selection by an explicit criterion matters here because a manifold method will produce a visually appealing picture at almost any setting. t-distributed stochastic neighbour embedding [7] is computed at a perplexity of 30 for visual contrast only, and is never used as an input to clustering, because its objective preserves local neighbourhoods at the expense of the global geometry a distance-based clustering algorithm depends upon.

Each club is summarised in two ways, as a minutes-weighted centroid of its outfield players in the shared principal component space and as the share of its outfield minutes played by each archetype, and clubs are then clustered hierarchically on archetype composition and compared against final league position. Player similarity is computed as cosine similarity within position group. Per-90 rates computed over partial seasons are noisy, and that noise is largest where the minutes are fewest, so we apply empirical Bayes shrinkage toward the position group mean with the degree of shrinkage determined by minutes played [8], refit the clustering on the shrunken features, and report how many assignments change; a partition that dissolves under shrinkage was substantially a description of sampling noise.

S2 Feature dictionaries and data availability

The two samples are described by different feature sets drawn from different pulls, so each has its own dictionary. Table S1 lists the 33 features of the primary sample with the FBref table each column was read from, and Table S2 lists the 18 features of the reduced sample, which combines what survived the withdrawal with an independent expected goals source. In both, counting statistics are expressed per ninety minutes played and rate features are used as published rather than divided a second time.

Every feature in the reduced set was confirmed to hold non-null values in both of its seasons before being admitted, which is the check that distinguishes a live column from one whose values were withdrawn. Table S3 reports that audit. The primary set is audited differently, by per-column and per-season coverage on load, because its failure mode is a block missing from one season rather than a column emptied across all of them.

Table S1: The primary feature set, with the FBref table each column was read from. Counting statistics are per ninety minutes played; the seven rate features are used as published. Five defensive counts are possession adjusted, with the elasticities in Section S3.

Feature	FBref table	Definition
np xg p90	standard	Non-penalty expected goals (per 90)
shots p90	shooting	Shots taken, excluding penalties (per 90)
goals non penalty p90	standard	Goals excluding penalties (per 90)
xag p90	standard	Expected assisted goals, the expected goals of shots a pass created (per 90)
key passes p90	passing	Passes leading directly to a shot (per 90)
sca p90	gca	Shot creating actions, the two offensive actions before a shot (per 90)
gca p90	gca	Goal creating actions, the two offensive actions before a goal (per 90)
passes final third p90	passing	Completed passes into the final third (per 90)
passes penalty area p90	passing	Completed passes into the penalty area (per 90)
crosses p90	misc	Crosses attempted (per 90)
progressive passes p90	passing	Completed passes moving the ball substantially toward goal (per 90)
progressive carries p90	possession	Carries moving the ball substantially toward goal (per 90)
progressive receptions p90	possession	Passes received in a progressive position (per 90)
carries final third p90	possession	Carries into the final third (per 90)
touches def pen p90	possession	Touches in the defending team's own penalty area (per 90)
touches def third p90	possession	Touches in the defensive third (per 90)
touches mid third p90	possession	Touches in the middle third (per 90)
touches att third p90	possession	Touches in the attacking third (per 90)
touches att pen p90	possession	Touches in the opposition penalty area (per 90)
take ons p90	possession	Take-ons attempted (per 90)
tackles padj p90	defense	Tackles, possession adjusted (per 90)
interceptions padj p90	misc	Interceptions, possession adjusted (per 90)
blocks padj p90	defense	Blocks of a pass or shot, possession adjusted (per 90)
clearances padj p90	defense	Clearances, possession adjusted (per 90)
ball recoveries p90	misc	Loose balls recovered (per 90)
fouls committed padj p90	misc	Fouls committed, possession adjusted (per 90)
shot accuracy pct	shooting	Shots on target as a share of shots
take on success pct	possession	Take-ons completed as a share of those attempted
tackle win pct	defense	Duels won as a share of those contested
aerials won pct	misc	Aerial duels won as a share of those contested
pass cmp short pct	passing	Short passes completed as a share of those attempted
pass cmp medium pct	passing	Medium passes completed as a share of those attempted
pass cmp long pct	passing	Long passes completed as a share of those attempted

Table S2: The outfield feature set. Counting statistics are per ninety minutes. The two rate features are used as published.

Feature	Source	Definition
np_xg_p90	Understat	Non-penalty expected goals
xa_p90	Understat	Expected assists
key_passes_p90	Understat	Passes leading directly to a shot
xg_chain_p90	Understat	Expected goals of possessions the player was involved in
xg_buildup_p90	Understat	As xGChain, excluding shots and key passes
shots_p90	FBref	Shots taken, excluding penalties
shots_on_target_p90	FBref	Shots on target
goals_non_penalty_p90	FBref	Goals excluding penalties
assists_p90	FBref	Assists
crosses_p90	FBref	Crosses attempted
interceptions_padj_p90	FBref	Interceptions, possession adjusted
tackles_won_padj_p90	FBref	Tackles won, possession adjusted
fouls_committed_padj_p90	FBref	Fouls committed, possession adjusted
fouls_drawn_p90	FBref	Fouls drawn
offsides_p90	FBref	Times caught offside
cards_yellow_p90	FBref	Yellow cards
shot_accuracy_pct	FBref	Shots on target as a share of shots, a rate
goals_per_shot	FBref	Goals per shot, a rate

Table S3: Populated columns in each FBref player table. The tables are still served with complete headers, so the empty counts are the number of columns that resolve correctly and contain no values.

Table	2024-25		2025-26	
	Columns	Empty	Columns	Empty
standard	17	0	17	0
shooting	11	0	11	0
passing	21	19	21	19
passing types	16	14	16	14
goal shot creation	17	16	17	16
defense	17	14	17	14
possession	21	20	21	20
playing time	17	0	17	0
misc	13	2	13	2
keeper	19	0	19	0
keeper adv	26	23	26	23

S3 Possession elasticities

Before any structure is estimated, the defensive counts have to be separated from the opportunity to record them. Team defensive volume does rise as opponent possession rises, so the confound is real, and the conventional correction, in the proportional form that has succeeded the original of Knutson [9], assumes the relationship is proportional: a side with twice the ball is taken to concede half the defensive events.

It is much weaker than that for 4 of the five statistics, and it is not distinguishable from proportional for the fifth. Fitted on the primary sample over 490 team-seasons, with 2,000 bootstrap resamples giving each estimate an interval, the elasticities are 0.172 for tackles (0.109 to 0.235), 0.353 for interceptions (0.171 to 0.540), 0.439 for blocks (0.379 to 0.499) and 0.443 for fouls committed (0.359 to 0.526). All four intervals exclude one, and no bootstrap resample of any of them reached it. Clearances are different: the estimate is 0.958, the interval runs from 0.823 to 1.091, 28.2 percent of resamples land at or above one, and the bootstrap probability that the true elasticity is at least one is 0.2975. For clearances the conventional exponent cannot be rejected, and the paper does not claim otherwise.

That pattern fits the football. A clearance is the most purely reactive action in the set, made when the ball arrives rather than sought, so it is the statistic for which the exposure story should hold best, and it is the one where proportionality survives. A tackle is the most volitional, and it is furthest from proportional. The spread from 0.172 to 0.958 is itself the point: a single exponent applied to every defensive statistic is wrong for four of them.

The reduced sample supplies the check the primary sample cannot, because it retains the team-level possession series needed to measure what is left after correcting. Across its 40 team-seasons, team possession ranges from 40.6 to 61.3 percent, and its fitted elasticities agree with the primary sample in falling well below one, at 0.525 for interceptions, 0.432 for tackles won and 0.283 for fouls committed. Table S4 shows what each choice of exponent leaves behind. The three raw correlations with team possession are -0.122, -0.111 and -0.102, all negative, since a player whose team keeps the ball records fewer defensive actions. Applying the conventional exponent overshoots every one of them, giving 0.070, 0.105 and 0.129, reversing sign in all three cases and, for fouls committed, ending further from zero than where it started. The correction passes through the confound and out the other side. Adjusting with the measured elasticity instead leaves -0.022, -0.018 and -0.037, three to four times closer to zero than the conventional exponent achieves.

Two qualifications belong with this. The residual-dependence check runs on one league, whose possession shares are relatively compressed, so the exact residuals should not be transported to a competition with a wider spread without refitting. Refitting the elasticities within each league tests transport directly, and the answer is mixed. For tackles, interceptions, fouls and clearances every league's interval overlaps the pooled one. Blocks do not transport: the league estimates run from 0.263 in La Liga to 0.615 in Premier League, and the intervals for both of those leagues miss the pooled interval entirely. Clearances are also the statistic where leagues disagree most, with 3 of the five league point estimates at or above one, which is a second reason not to claim sub-proportionality for them.

The constant elasticity form is itself an assumption, and for 3 of the five statistics it fails. Adding a quadratic term in log opponent possession improves the fit for tackles, blocks and fouls, with a coefficient whose interval excludes zero in each case, and in each the response flattens toward zero for the sides that see least of the ball. Fouls committed bend most: the implied elasticity is 0.90 at 37.6 percent opponent possession and -0.07 at 58.9 percent. The exponents are therefore averages over the possession range rather than constants. Figure S1 gathers the pooled estimates, their intervals and the league refits, and places both families of statistic either side of the proportional line.

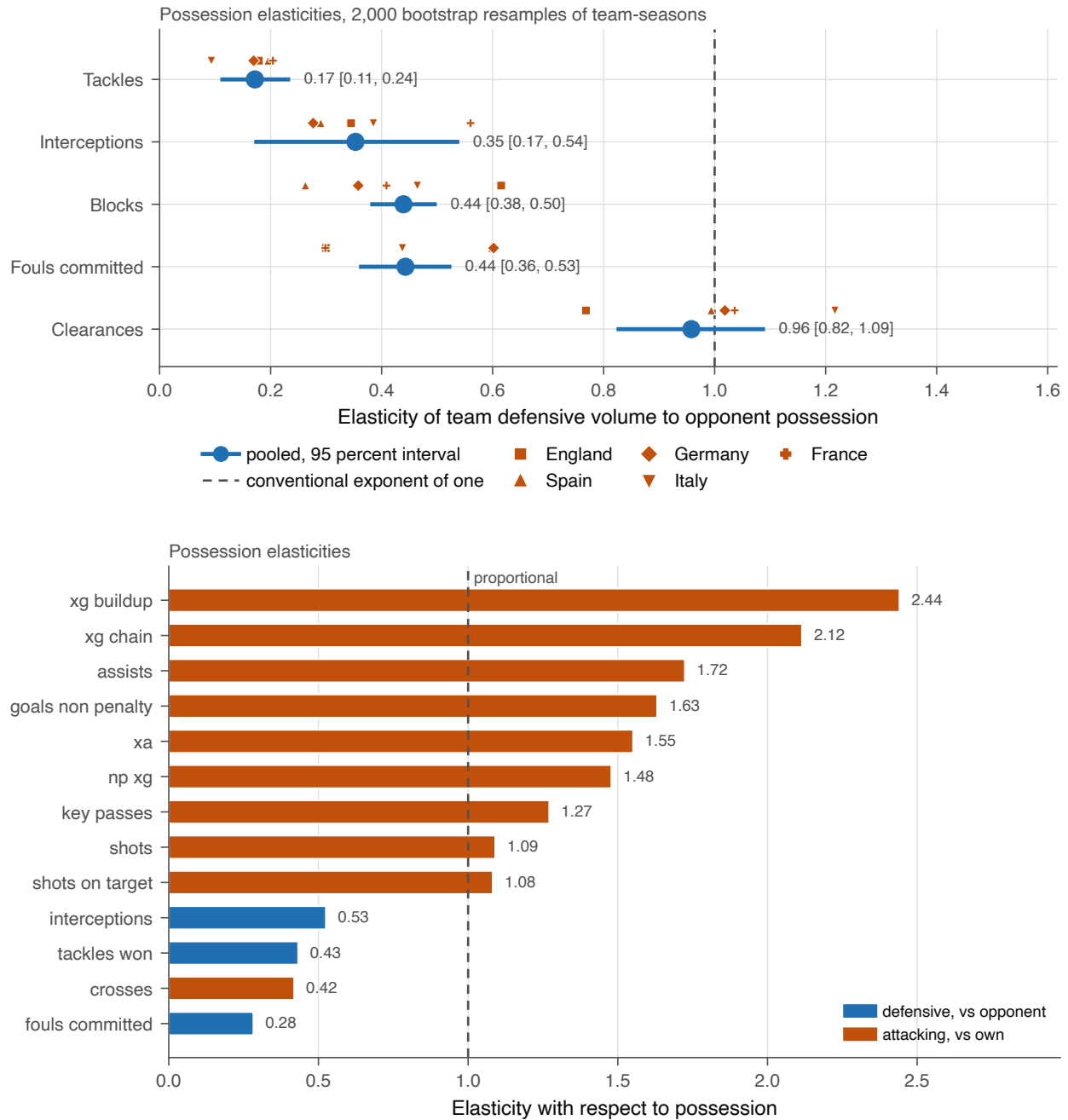


Figure S1: Possession elasticities. Top, the five defensive elasticities with bootstrap intervals from 2,000 resamples of 490 team-seasons, against the conventional value of one, with each league's own refit shown as a smaller marker. Bottom, both families of statistic on the reduced sample: defensive volume responds to opponent possession less than proportionately, while attacking output responds to a team's own possession more than proportionately.

Table S4: Possession adjustment. The elasticity is fitted from team totals. The final three columns give the correlation between each per-90 rate and team possession: unadjusted, adjusted with the fitted elasticity, and adjusted with the conventional exponent of one. A value near zero is the goal.

Statistic	Elasticity	Correlation with team possession		
		Unadjusted	Adjusted	Exponent one
Interceptions	0.525	-0.122	-0.022	0.070
Tackles won	0.432	-0.111	-0.018	0.105
Fouls committed	0.283	-0.102	-0.037	0.129

S3.1 Why the correction is one sided

Correcting only the defensive statistics invites an obvious objection. Attacking output also depends on possession, so the feature set is still team dependent and the correction looks like half a correction. The objection deserves a test rather than an assertion. The test runs on the reduced sample, which is the only one of the two carrying an independent expected-goals source with involvement measures, and its conclusion concerns the shape of the two relationships rather than their exact magnitudes.

The two sides of the ball do not behave alike. Fitted the same way, against a team’s own possession rather than the opponent’s, the attacking elasticities are 1.09 for shots, 1.48 for non-penalty expected goals, 1.63 for non-penalty goals, 1.72 for assists, 2.12 for xGChain and 2.44 for xGBuildup. Every one except crossing, at 0.42, exceeds one, while 4 of the five defensive elasticities fall well below it and the fifth cannot be distinguished from it.

That asymmetry is the whole argument. For defending, opponent possession is an exposure denominator in the plain sense that a tackle cannot occur while your own side holds the ball, and the sub-proportional response says exposure is what it measures. For attacking, an elasticity above one says the opposite: ten percent more of the ball buys appreciably more than ten percent more output. Returns to possession are increasing, which is a statement about football rather than about measurement, so dividing it out would remove signal along with any bias. The two involvement measures make the point most starkly, since both exceed two.

The counterfactual was nonetheless computed. Applying the fitted exponents to the attacking block as well moves the correlation between club centroids and final league position from -0.861 to -0.610, so a symmetric correction does remove a real part of the team relationship rather than nothing at all. It does not remove all of it, the partition moves at an adjusted Rand index of 0.872 against the one reported here, and the silhouette drifts from 0.285 to 0.281. The choice is a judgement rather than a fact, and it is recorded as one: we treat possession driven productivity as something a role profile should reflect, and a reader who takes the opposite view has the numbers to see what that would cost.

S4 Feature-family ablation in full

Table S5 gives every cell of the leave-one-family-out and keep-one-family-only ablation summarised in Section 3.4: for each feature set and each scope, the number of clusters the declared rule selects, the standard score of the observed two-cluster silhouette against the clusterless null, and the adjusted Rand index between the ablated partition and the full one on the same players. The last of those is what distinguishes a surviving count of two clusters from the same two modes surviving.

Table S5: Standard score of the observed two-cluster silhouette against the clusterless Gaussian null, by scope, under every feature-family ablation. A dagger marks a cell where the pre-declared rule chose a number of clusters other than two.

Feature set	All	DF	MF	FW
<i>Full feature set</i>				
all 33 features	18.0	24.2	15.0	13.2
<i>Leave one family out</i>				
without shooting	17.7	21.3	12.3	18.0
without creation	21.2	21.1	14.5	15.1
without progression	19.0	19.7	13.2	19.8
without territory	21.6	22.6	16.0	15.6
without defending	19.7	29.1	16.0	17.2
without passing	19.1	17.5	10.2	20.1
<i>One family alone</i>				
shooting only	29.3	19.8	16.1	2.9
creation only	12.8	22.6	11.7	18.4
progression only	8.3	15.6	9.0	12.9
territory only	11.4	26.6	16.2	20.2
defending only	19.0	6.9	10.7	8.5
passing only	6.6	11.4	5.3	1.4

S5 Nesting: resampling players rather than player-seasons

The unit of analysis is the player-season, and 9,263 of them come from 3,475 players, so an average player supplies 2.7 rows. A bootstrap that resamples rows therefore treats correlated observations as independent, and the stability it reports is optimistic by an unknown amount. This section measures the amount.

The identical bootstrap is run twice on the same partition with the same number of resamples, 50. The first draws player-seasons with replacement, which is what the rest of the paper reports and what the genre does. The second draws whole players with replacement and takes every season a drawn player contributed, so that rows sharing a player are drawn or left out together. The adjusted Rand index is computed the same way in both, on the distinct rows the draw contains.

The difference is small. On the outfield pool the row bootstrap returns 0.988 and the player block bootstrap 0.980. Within position groups the block index is 0.983 for defenders, 0.974 for midfielders and 0.938 for forwards, against 0.990, 0.979 and 0.962 for the row bootstrap. The largest fall anywhere is 0.024, for forwards, which is the smallest scope and the one the paper already reports as the least well supported.

Two qualifications belong with the comparison. A player block bootstrap draws fewer distinct rows than a row bootstrap of the same nominal size, because one drawn player contributes several correlated rows at once, so a small fall is expected from the reduced effective sample alone and is not by itself evidence that the row bootstrap was misleading. The check also applies to the observed data only: the clusterless datasets of Section 2.3 are independent rows by construction, with no player to block on, so the comparison between observed and null stability is unaffected.

What the check establishes is narrow and worth having. The stability reported for this partition does not depend on treating player-seasons as exchangeable, so the argument that stability fails to distinguish real structure from a partitioned continuum is not an artefact of the resampling unit either.

Table S6: Sensitivity of the analysis to the minutes threshold in the earlier of the two reduced-sample seasons. Agreement is the adjusted Rand index against the partition at the threshold actually used, computed on the players common to both pools.

Minutes	Eligible	k	Silhouette	Bootstrap ARI	Agreement
270	403	2	0.286	0.930	0.967
450	366	2	0.286	0.924	baseline
900	296	2	0.302	0.947	0.986

S6 Minutes threshold sensitivity

Both samples exclude outfield players below 450 minutes. That threshold balances two errors, since a low threshold admits players whose per-90 rates are dominated by sampling noise while a high threshold discards genuine squad roles such as the impact substitute. Table S6 repeats the entire selection and stability procedure at 270, 450 and 900 minutes, treating each threshold as an independent pool in which imputation and standardisation are recomputed and the declared rule is reapplied from scratch.

The threshold matters very little. The rule selects two clusters at every threshold, the silhouette moves only from 0.286 to 0.302, and bootstrap stability is between 0.924 and 0.947 throughout. The partition itself barely moves: measured on the players common to both pools, the adjusted Rand index against the partition at the threshold actually used is 0.967 at 270 minutes and 0.986 at 900. Figure S2 plots the same quantities.

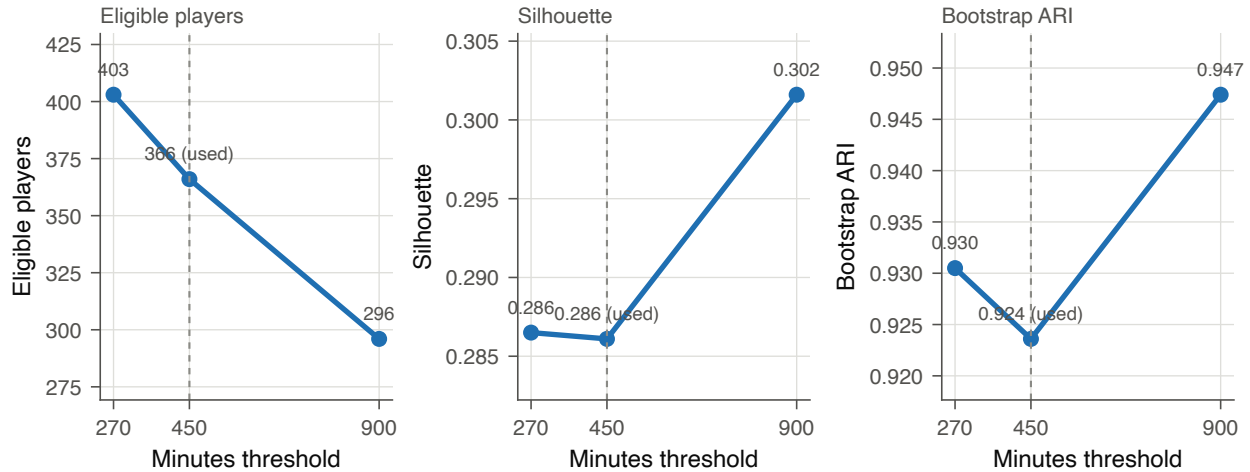


Figure S2: Eligible sample, silhouette and bootstrap stability across the candidate minutes thresholds. The dashed line marks the threshold used in the main analysis.

S7 Selection criteria and embedding choices

Table S7 gives the silhouette coefficient, Davies-Bouldin index, Calinski-Harabasz index and gap statistic for every candidate number of clusters on the reduced sample, for the global analysis and within each position group, with the row the declared rule selects marked in each scope. The corresponding evidence for the primary sample is Figure 1, which carries more information than a

criteria table would, because the question the primary sample answers is not which candidate scores best but whether any of them beats what a continuum produces.

Table S8 reports trustworthiness for all twelve configurations of the neighbour count and minimum distance, on the reduced sample, which is the only one for which a manifold embedding is reported. Selection was by trustworthiness alone, decided before any embedding was inspected visually, which matters because a manifold method will produce a visually appealing picture at almost any setting.

Table S7: Cluster selection criteria for every candidate number of clusters, by scope. The consensus rule is applied to these values.

Scope	k	Silhouette	Davies-Bouldin	Calinski-Harabasz	Gap
global*	2	0.287	1.451	145.6	1.216
global	3	0.215	1.695	101.9	1.283
global	4	0.154	1.954	86.1	1.339
global	5	0.145	1.861	75.4	1.381
global	6	0.144	1.797	68.3	1.415
global	7	0.135	1.905	61.3	1.438
global	8	0.118	1.881	56.1	1.458
global	9	0.123	1.840	53.5	1.477
global	10	0.123	1.805	49.8	1.484
global	11	0.128	1.657	47.6	1.506
global	12	0.121	1.911	44.6	1.517
DF*	2	0.204	2.037	25.8	0.943
DF	3	0.151	1.919	23.5	1.011
DF	4	0.142	1.983	19.7	1.036
DF	5	0.120	1.961	18.0	1.059
DF	6	0.117	1.795	16.9	1.083
DF	7	0.106	1.898	15.6	1.109
DF	8	0.120	1.792	14.7	1.136
DF	9	0.120	1.754	14.1	1.154
DF	10	0.112	1.636	13.5	1.159
DF	11	0.098	1.636	12.8	1.162
DF	12	0.105	1.627	12.3	1.196
MF*	2	0.243	1.543	69.4	1.035
MF	3	0.147	1.969	49.0	1.091
MF	4	0.145	1.895	40.4	1.130
MF	5	0.145	1.873	34.7	1.153
MF	6	0.132	1.939	30.3	1.167
MF	7	0.122	2.054	27.4	1.199
MF	8	0.107	2.030	24.9	1.202
MF	9	0.103	1.923	23.8	1.212
MF	10	0.106	1.983	21.9	1.220
MF	11	0.094	1.944	20.4	1.227
MF	12	0.097	1.835	19.5	1.226
FW*	2	0.187	1.803	12.4	0.764
FW	3	0.175	1.576	10.8	0.837

Table S8: UMAP hyperparameter grid, ranked by trustworthiness. Selection was by trustworthiness alone and was decided before any embedding was inspected.

Neighbours	Minimum distance	Trustworthiness
10	0	0.9322
10	0.10	0.9316
15	0	0.9306
15	0.10	0.9303
30	0.10	0.9276
30	0	0.9257
10	0.50	0.9186
15	0.50	0.9175
50	0.10	0.9169
50	0	0.9156
30	0.50	0.9083
50	0.50	0.9045

Scope	k	Silhouette	Davies-Bouldin	Calinski-Harabasz	Gap
FW	4	0.176	1.599	10.2	0.908
FW	5	0.196	1.181	9.3	0.909
FW	6	0.124	1.490	8.5	0.944
FW	7	0.132	1.350	7.8	1.017
FW	8	0.101	1.389	7.7	1.021
FW	9	0.127	1.229	7.7	1.070
FW	10	0.125	1.350	7.8	1.051
FW	11	0.128	1.199	7.1	1.064
FW	12	0.116	1.126	6.8	1.096

S8 Fuller views of the main-text figures

Figure S3 gives the three panels of Figure 2 at full size and with the detail a six-panel composite cannot carry: the principal component loadings across all five leading components rather than the first two, the mode against position contingency normalised by row as well as by column, and the confusion matrices for both classifiers rather than only the boosted one. Figure S6 gives the centroid deviation of each archetype from its position group mean across the entire feature set, rather than only the ten features with the largest separation shown in Figure 3. Figure S7 gives the same centroids as radars on fixed axes within each position group. Within a group the two profiles are mirror images by construction, since the features are standardised within group before clustering, so the useful reading is of shape rather than of extent.

Figure S9 gives the full feature attribution for the position classifier. The direction of each attribution adds a detail that magnitude alone hides. Ranking the correlation between a feature’s value and the attribution it makes to a class separates a class identified by what its members do from one identified by what they do not, and the three classes differ on exactly that. 4 of the five strongest features for defenders are positively signed, so defenders are recognised by the presence of what they do. Forwards and midfielders are recognised as much by the absence of defensive volume as by the presence of attacking output, with only 2 and 2 of their five positive respectively. A class

defined largely by negations is a class occupying one end of a scale rather than a category with a signature of its own. Figure S10 splits the same attribution by class.

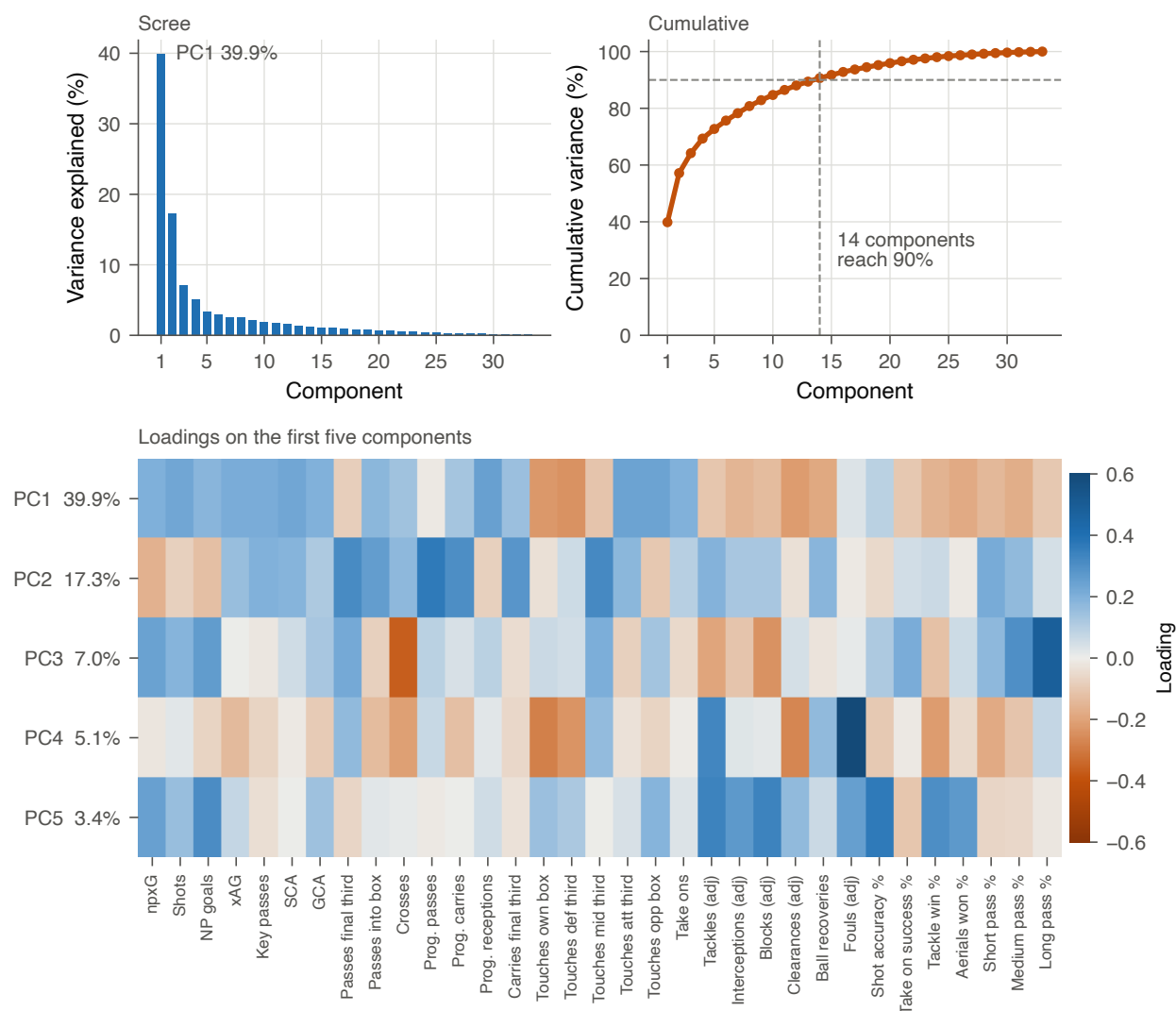


Figure S3: Principal components of the 33 archive features over 9,263 player-seasons, with the scree curve and the loadings of the leading components at full size. Figure S4 gives the contingency and Figure S5 the confusion matrices.

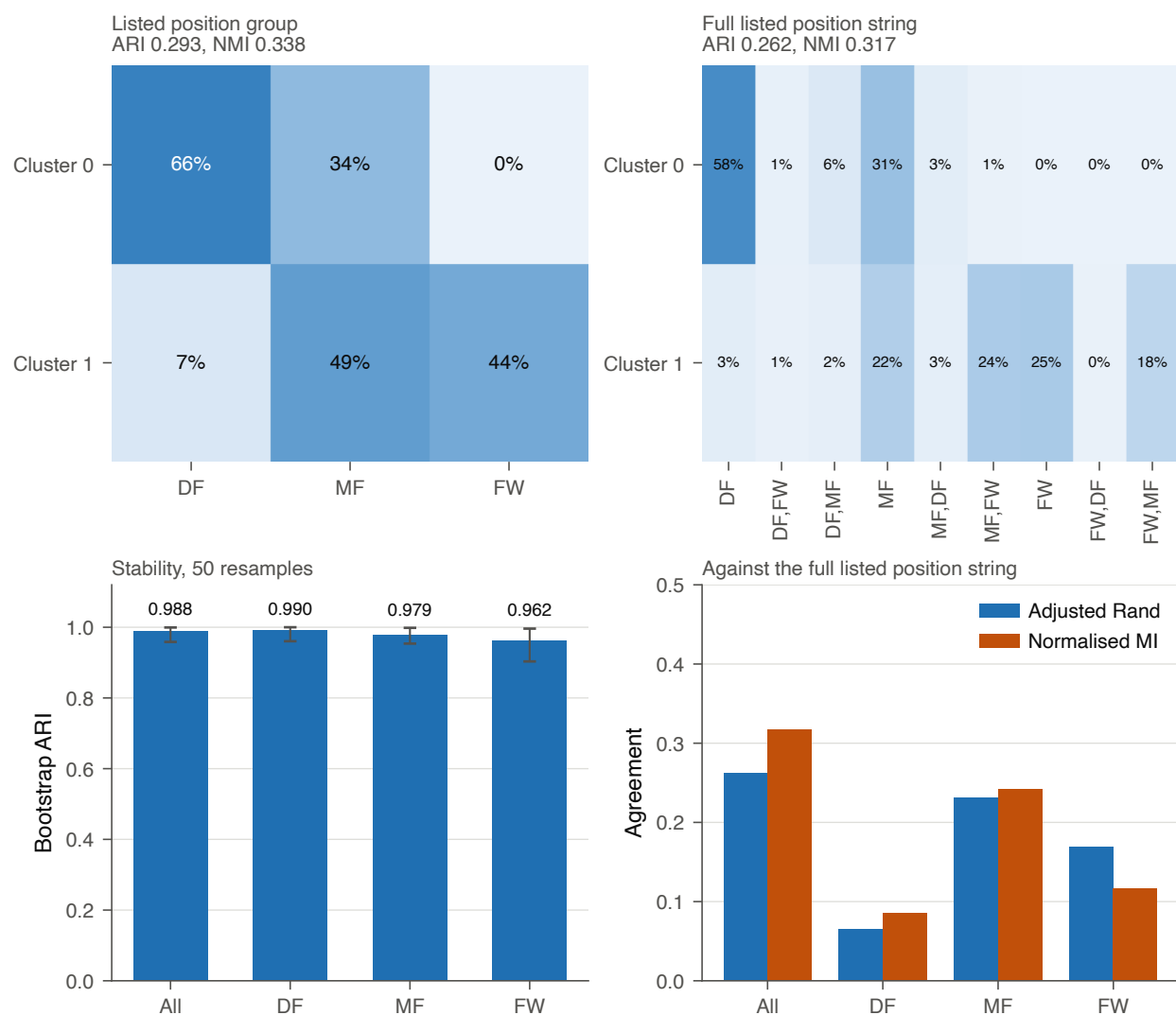


Figure S4: The pooled two-mode partition against listed position, with the heat maps normalised by row so that each gives the position composition of one mode. The main text reads the same table by column, which is where the 92.9 and 99.8 percent figures come from.

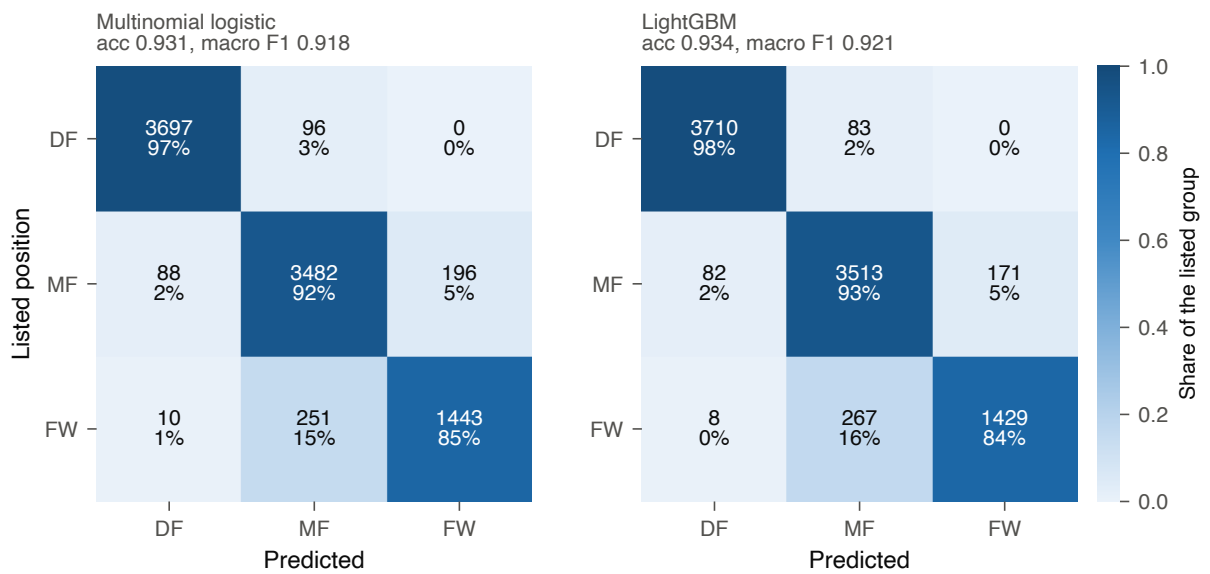


Figure S5: Out-of-fold confusion for both position classifiers on 9,263 player-seasons. The near equality of the linear and the boosted model is itself evidence that little interaction structure remains for the trees to find.

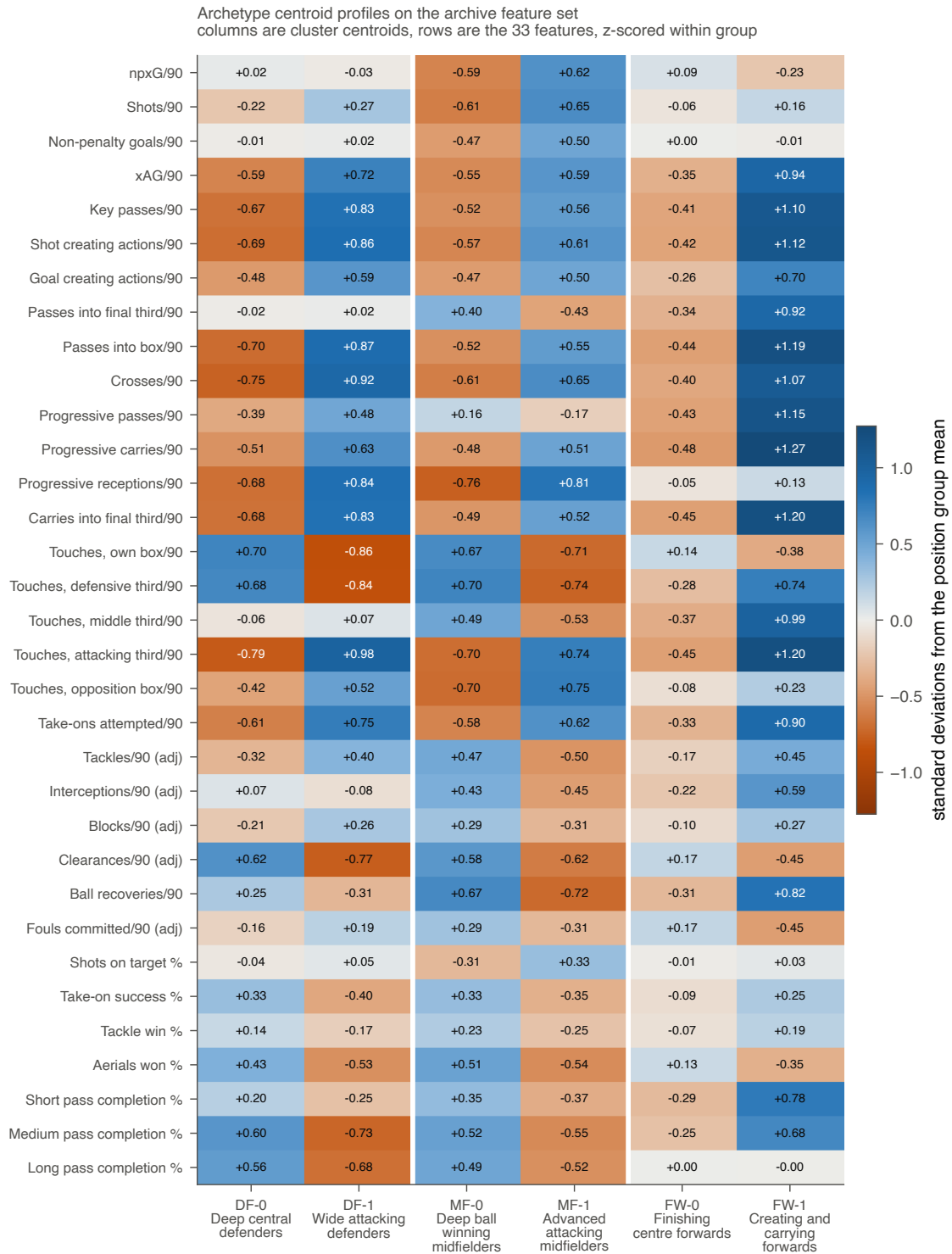


Figure S6: Centroid deviation of each archetype from its position group mean across the full 33 feature set, primary sample. Within a group the two profiles are mirror images by construction.

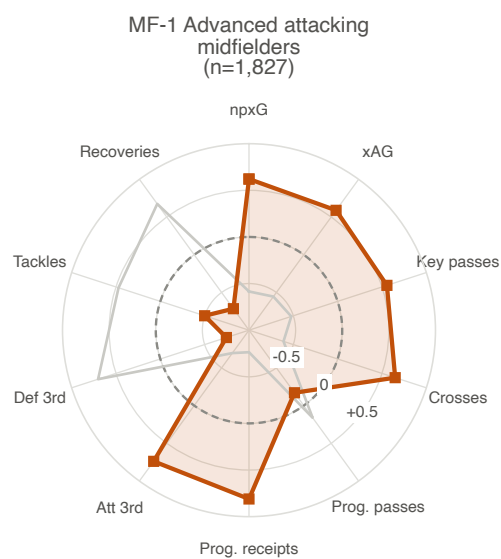
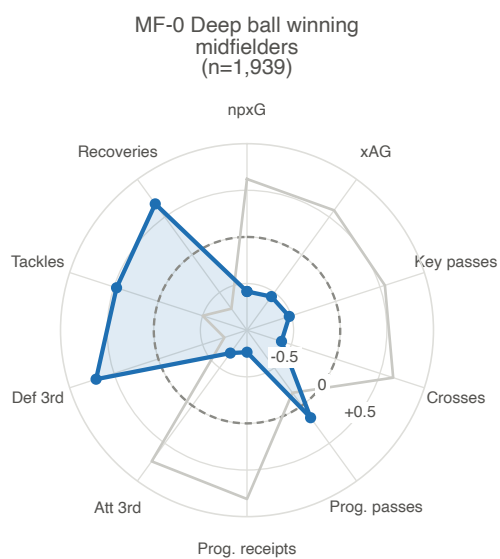
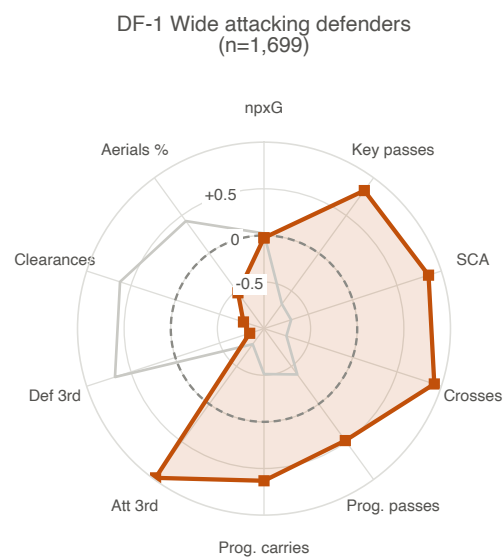
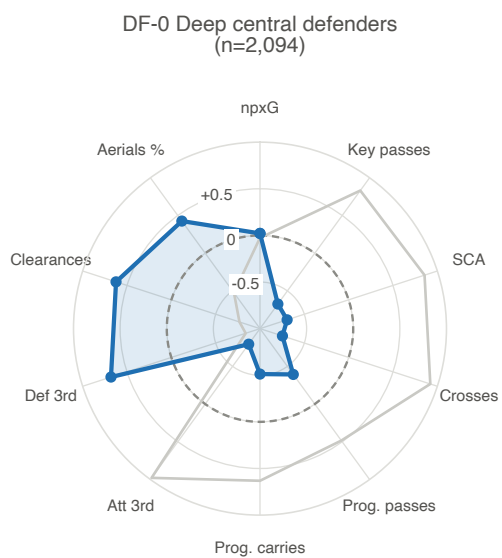


Figure S7: Archetype radars on fixed axes, primary sample. Top, defenders. Bottom, midfielders. Forwards are in Figure S8.

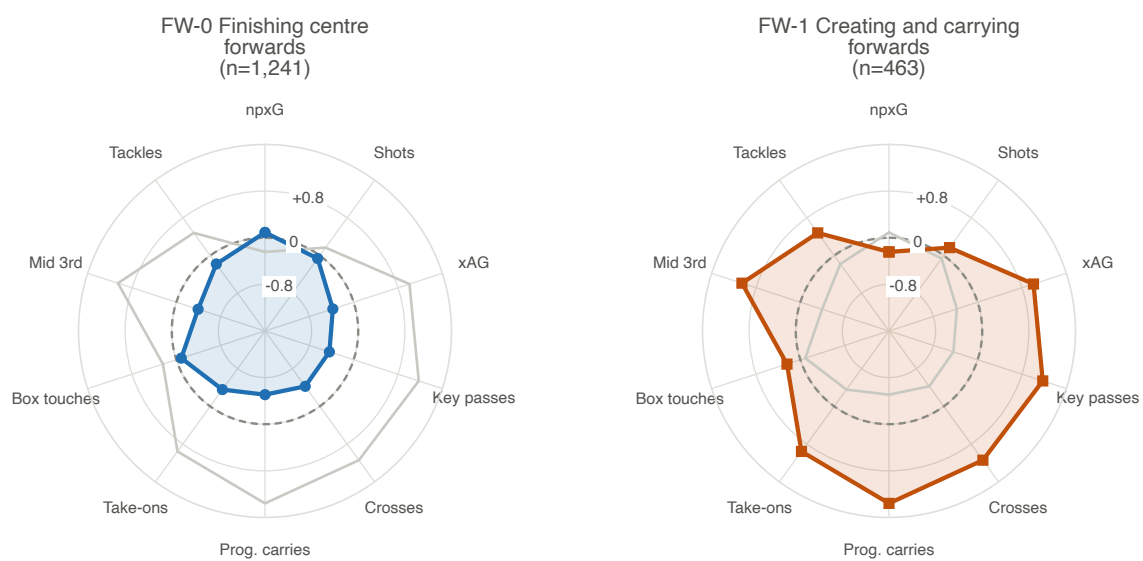


Figure S8: Archetype radars for forwards on the primary sample, on the same fixed axes as Figure S7.

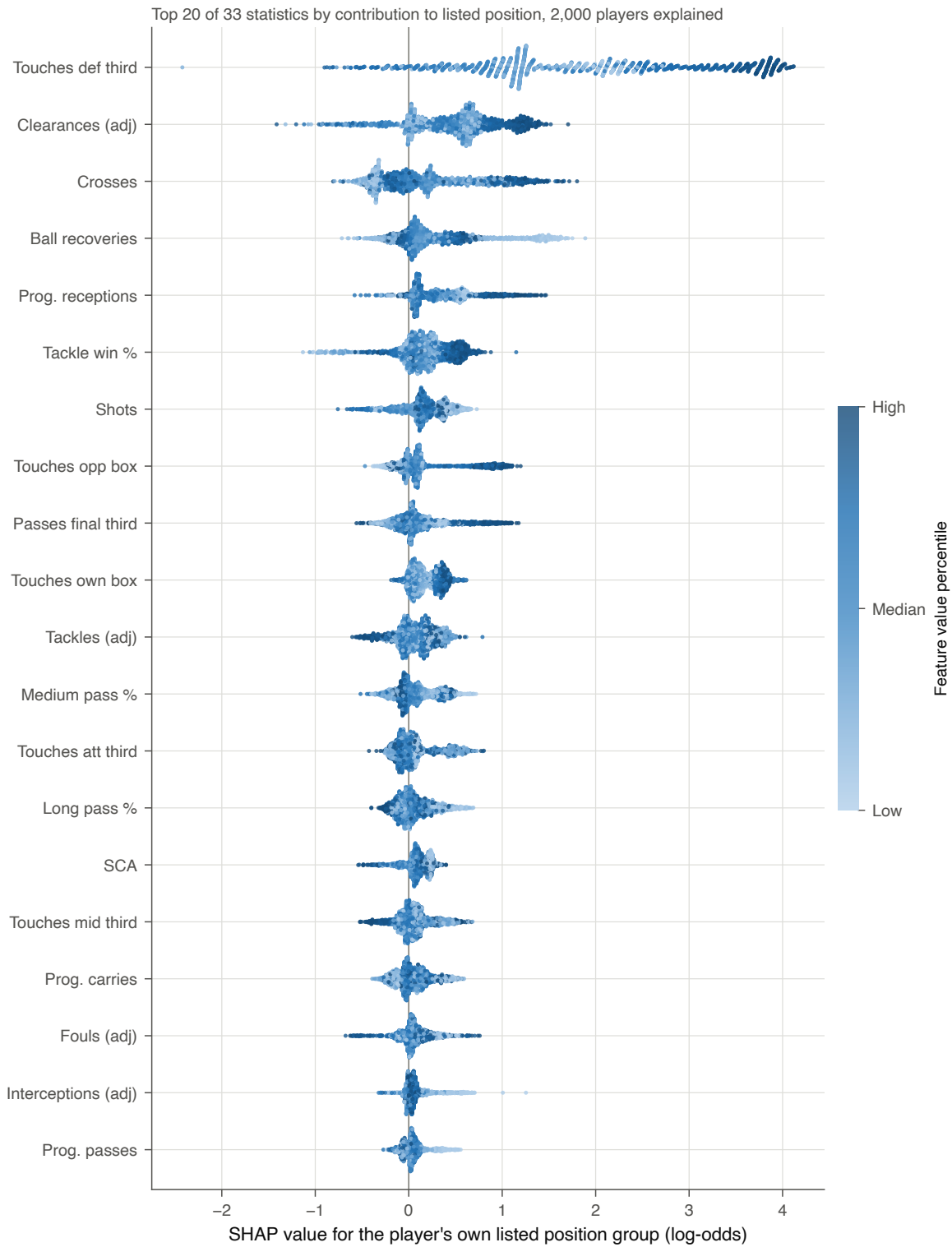


Figure S9: SHAP values for the gradient boosted position classifier on the primary sample, showing for each player the contribution of each feature toward the position group the player is actually listed at. Territorial features dominate technical ones throughout.

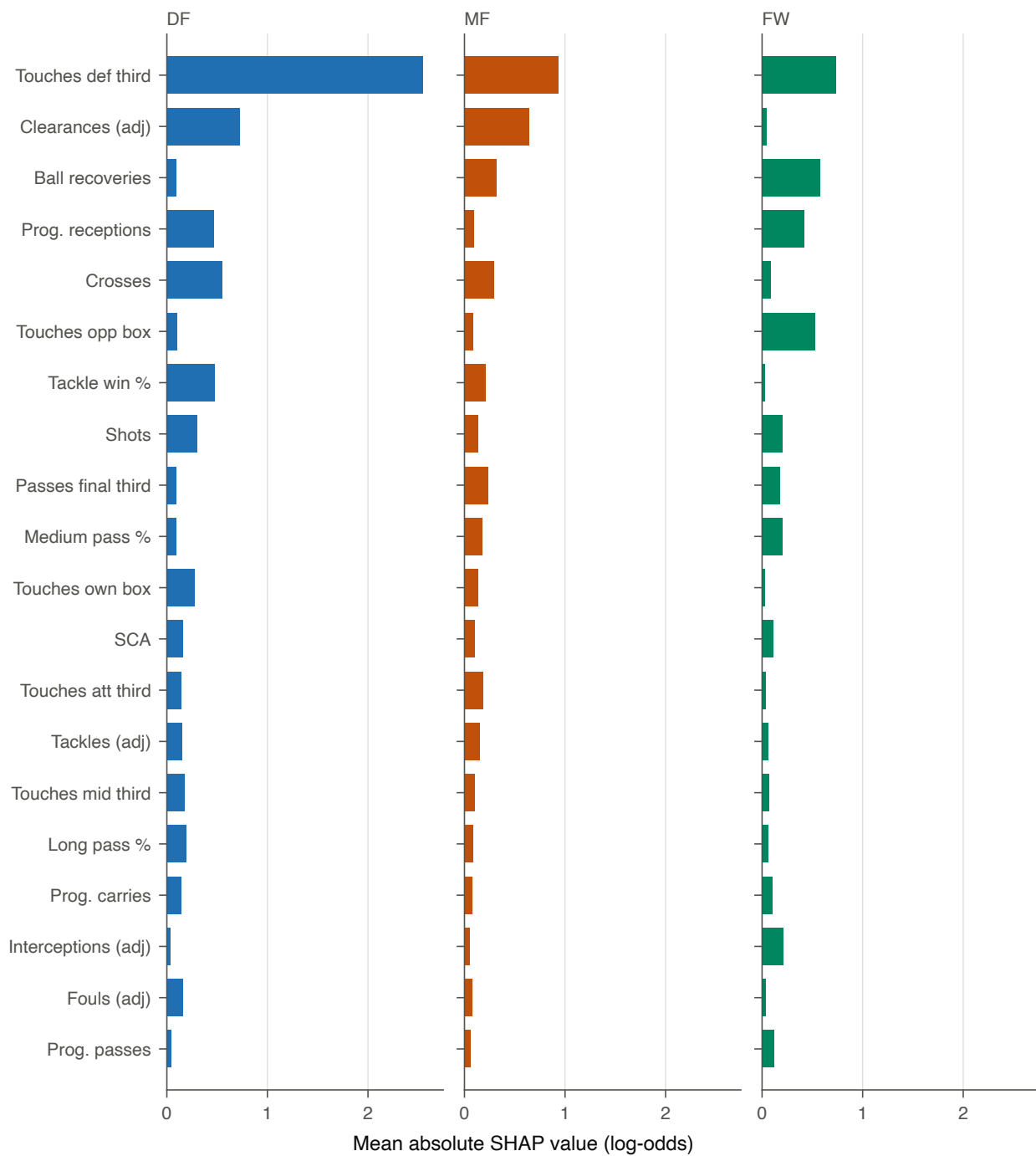


Figure S10: Mean absolute attribution per feature, split by position group, primary sample. Touches in the defensive third dominate for all three, and the profile of what matters differs sharply between them: defenders are separated largely by defensive volume and crossing, forwards by where they receive the ball. The panel shows magnitude only, and the direction each feature pushes is reported in the text.

S9 Players the description fits worst

A two-mode description of a continuum will fit some players badly, and which players those are is a test of whether the description is measuring football or noise. Isolation forests and Mahalanobis distance under a shrinkage covariance were fitted separately within each position group, and the two detectors agree on 80.5 percent of the players they flag.

The 15 strongest outliers are 12 distinct players, headed by Neymar. Two features of the list stand out. 11 of the 15 carry a compound listed position such as midfielder and forward together, against a much smaller share in the pool, so the players the partition handles worst are the ones the listing system also declines to place. The detector is also not simply finding stars: it finds attacking full backs whose output profile resembles a winger's, which is the same boundary the defender split describes and a genuine tactical category rather than an artefact.

One caveat belongs with the list rather than after it. 9 of the 15 played under nine hundred minutes in the season concerned, so their per-90 rates carry more sampling noise than a full season's would, and extreme rates are easier to reach on a small denominator. The list should be read as evidence about the kind of player the two-mode description struggles with, not as a ranking in which the fifteenth entry means what the first does. Figure S11 shows both detectors against each other.

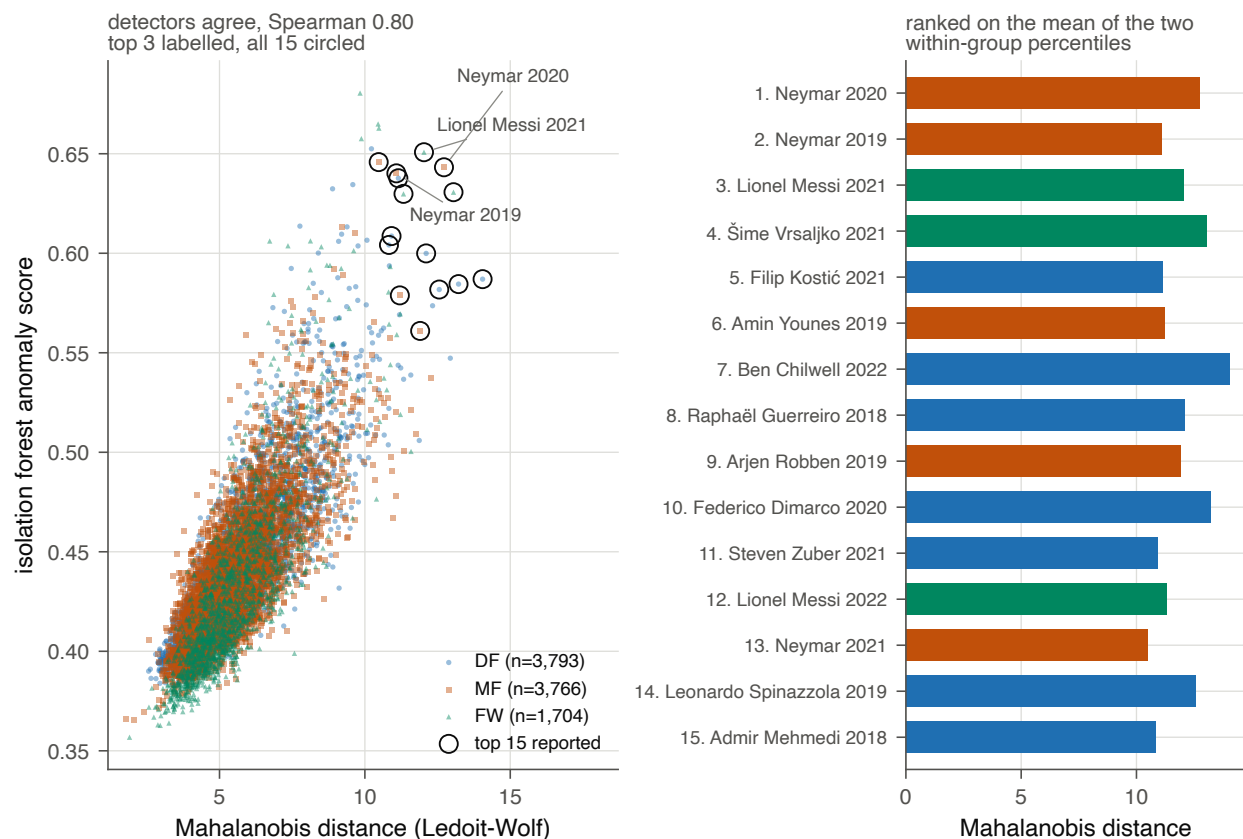


Figure S11: Isolation forest score against Mahalanobis distance within position group, primary sample. The two detectors are computed independently and rank the same players highly, and the flagged points are dominated by players carrying compound listed positions.

Table S9: The ten feature pairs whose Spearman correlation differs most across position groups. Every pair changes sign between at least two groups.

Feature A	Feature B	DF	MF	FW	Spread
Shots on target	Crosses	-0.01	0.43	-0.44	0.87
Crosses	Shot accuracy %	-0.22	0.26	-0.57	0.82
Crosses	Interceptions (adj)	0.21	-0.50	0.30	0.81
npG	Crosses	-0.15	0.36	-0.43	0.79
Key passes	Tackles won (adj)	0.47	-0.32	0.26	0.78
Crosses	Tackles won (adj)	0.51	-0.25	0.48	0.76
xA	Shots on target	0.17	0.56	-0.19	0.75
Shots	Interceptions (adj)	0.22	-0.50	-0.04	0.72
xA	Tackles won (adj)	0.39	-0.30	0.15	0.69
xA	Interceptions (adj)	0.19	-0.50	0.16	0.69

S10 Statistics that change meaning by position

Correlations were computed within each position group and compared across groups, and the ten pairs whose correlation varies most between groups are reported in Table S9 and shown in Figure S12. The largest spread between position groups reaches 0.87, against a median spread across all pairs of 0.38.

Every one of the ten most divergent pairs changes sign between position groups. The relationship between shots on target and crosses is positive for midfielders and negative for forwards, which is to say that a midfielder who crosses often is generally a more active attacker while a forward who crosses often is generally not the player getting on the end of chances. The relationship between key passes and tackles won is positive for defenders and negative for midfielders, distinguishing the full back whose defensive work and creative output rise together from the midfield division of labour between a destroyer and a creator.

The practical implication is that a single pooled correlation matrix, of the kind that appears routinely in public football analysis, averages over these sign reversals and reports a number that describes no actual population of players. It also supplies the justification for the position stratified analysis used throughout this paper, since a global embedding fitted across all outfield players is fitted across relationships that point in opposite directions. The full correlation matrix the comparison is drawn from is Figure S13.

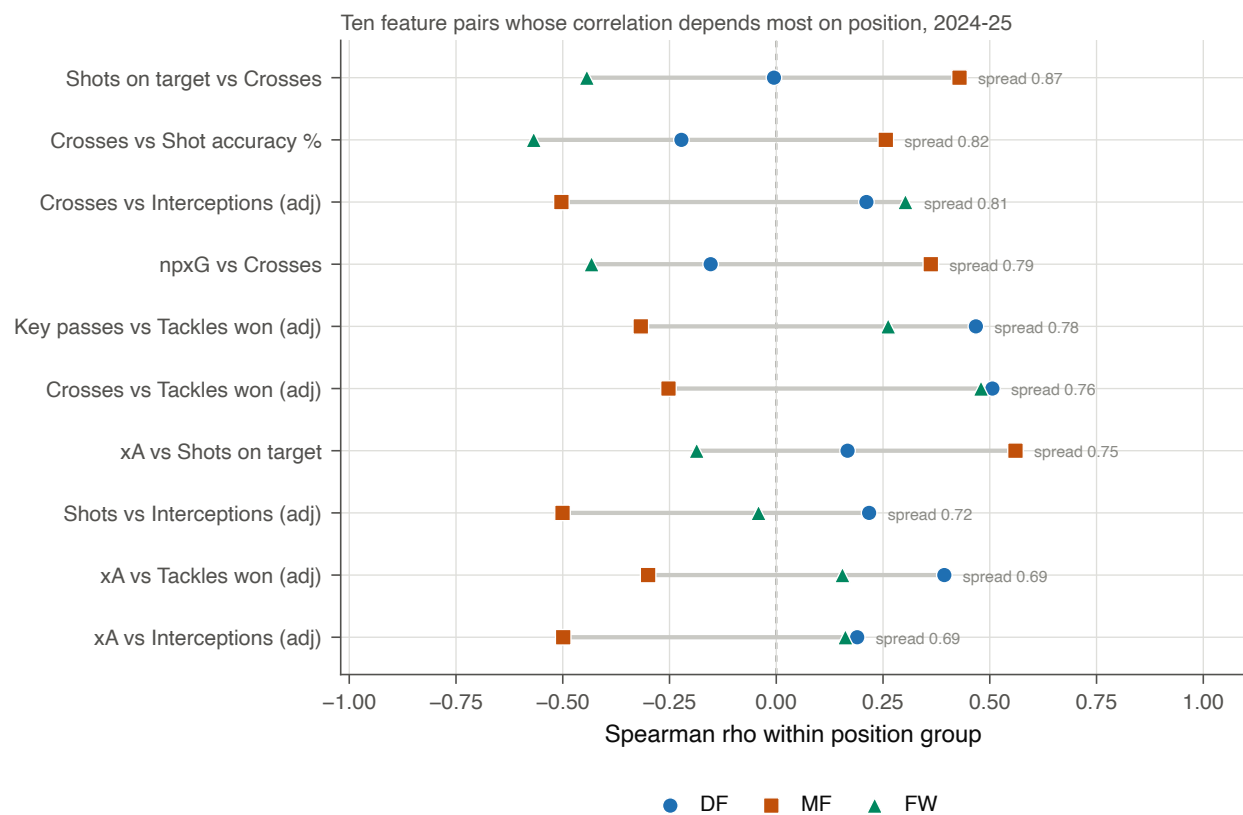


Figure S12: The ten feature pairs whose within-group correlation differs most across position groups, in the earlier of the two reduced-sample seasons. Every pair changes sign between at least two groups.

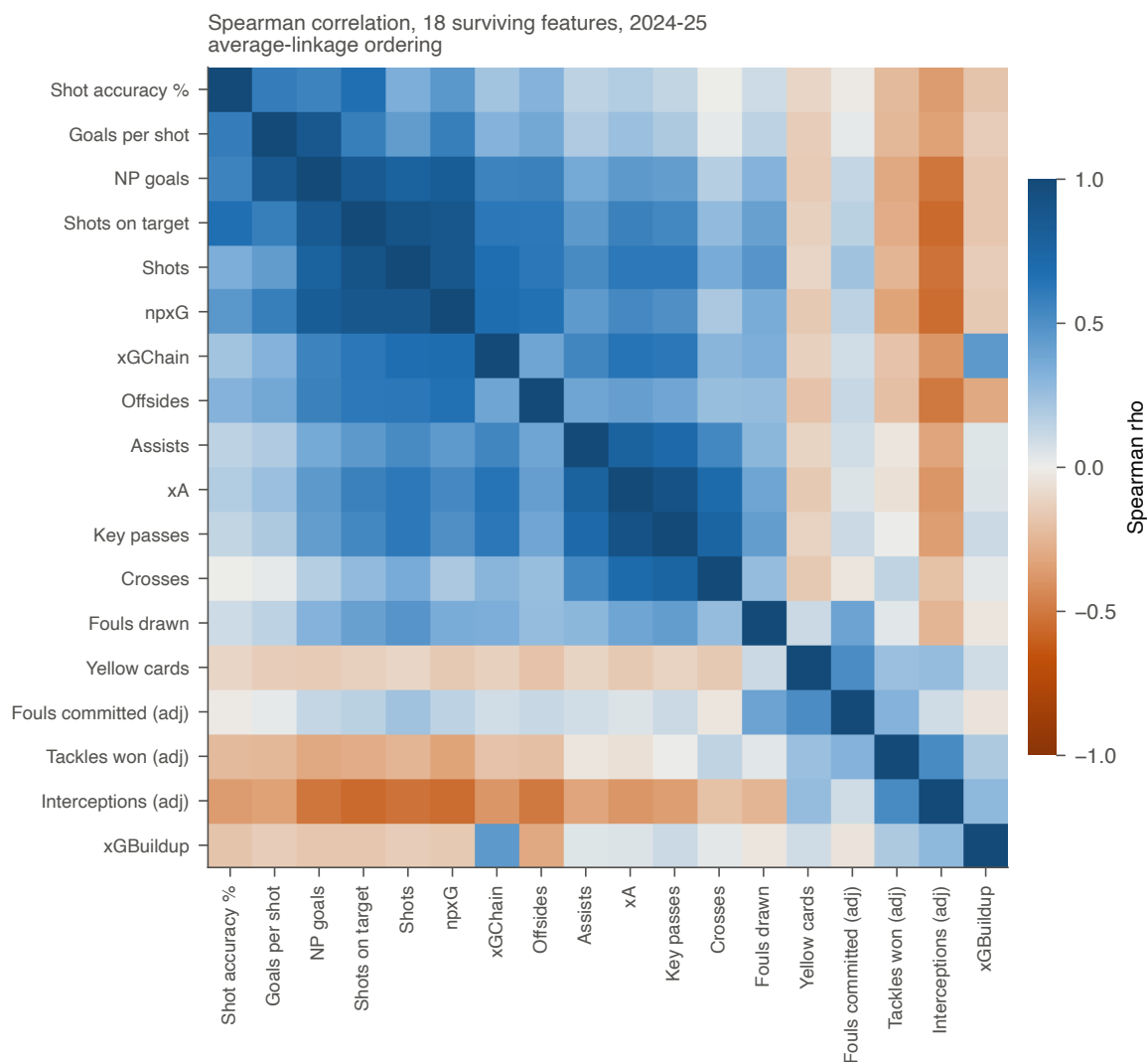


Figure S13: The full hierarchically clustered Spearman correlation matrix over the reduced outfield feature set, from which the comparison in Figure S12 is drawn.

S11 Analyses on the reduced sample

The analyses collected here run on the reduced sample described in Section 2.1: the seasons ending 2025 and 2026, pulled live after the withdrawal, with roughly half the features of the primary sample.

One point about scope needs stating plainly, because it applies to every figure in this section. Where a figure here shows Premier League players only, that is not a sampling choice made for convenience. It is the coverage of the live post-withdrawal pull that this robustness case is built on, and the reduced sample exists to test whether the main result survives a much poorer description of the same game rather than to characterise European football. The paper’s claims rest on the primary sample, which covers all five leagues and all five seasons, and every one of them is refitted independently in each league and each season in Section 3.4. Section S11.3 below additionally repeats the reduced-sample pipeline across all five leagues, so the reduced sample’s own generality is tested rather than assumed. The within-season drift analysis of Section S11.7 is narrower still, covering one club, and the reason is stated there.

S11.1 Cluster selection and stability

Applying the declared rule to the four selection criteria gives $k = 2$ globally and within each position group, matching the primary sample. Across 200 bootstrap resamples the mean adjusted Rand index against the reference partition is 0.914 globally and 0.815 for midfielders, both describing a partition that reappears almost unchanged under resampling. For defenders the mean falls to 0.607 with a standard deviation of 0.216, and for forwards to 0.602 with a standard deviation of 0.281. The minimum adjusted Rand index observed across forward resamples is -0.065, which is to say that in the least favourable resample the partition recovered is no better than chance. Forwards number only 49 players above the minutes threshold here, so this is partly a sample size problem, and it is a real limit on what the reduced sample can claim. The global partition scores an adjusted Rand index of 0.226 against the position group with a cluster purity of 0.579, so the clusters are not a rediscovery of the positions the source already records. Figure S19 gives the criterion curves, Figure S20 the co-assignment structure and Figure S21 the correspondence with position.

S11.2 Descriptive structure and embeddings

After filtering, 366 outfield players are eligible in the earlier of the two reduced-sample seasons, comprising 136 defenders, 181 midfielders and 49 forwards. Summary statistics by position group appear in Table S10. The feature set is strongly non-normal: all 18 of the 18 features reject the Shapiro-Wilk null at the five percent level, mean absolute skewness is 1.13, and Mardia’s test rejects multivariate normality decisively on both components. Principal component analysis requires no distributional assumption to be computed, so this does not invalidate the decomposition, but the components should be read as directions of variance rather than as latent factors, and it justifies the use of Spearman rather than Pearson correlation throughout.

Principal component analysis, UMAP and t-SNE were fitted to the same standardised features. The selected UMAP configuration used 10 neighbours at a minimum distance of 0. Trustworthiness is 0.855 for the linear projection, 0.932 for UMAP and 0.944 for t-SNE, while continuity is close to identical across the three. Nearest neighbour position purity at ten neighbours is 0.649 for the linear projection and 0.699 for UMAP, so roughly two thirds to three quarters of a player’s closest statistical neighbours share his listed position group. That figure is informative in both directions: it confirms the surviving features carry real positional signal, and it establishes that a substantial minority of every player’s nearest neighbours are listed at a different position, which is the space in which data-driven roles can differ from listed ones. All three embeddings display a continuous

Table S10: Headline per-90 statistics by position group in the earlier of the two reduced-sample seasons, reported as mean with standard deviation in parentheses.

Statistic	DF	MF	FW
np_xg_p90	0.05 (0.04)	0.16 (0.12)	0.41 (0.17)
xa_p90	0.06 (0.06)	0.15 (0.10)	0.16 (0.11)
key_passes_p90	0.51 (0.44)	1.26 (0.63)	1.05 (0.54)
xg_chain_p90	0.35 (0.16)	0.50 (0.20)	0.63 (0.20)
xg_buildup_p90	0.30 (0.14)	0.28 (0.13)	0.18 (0.09)
shots_p90	0.54 (0.33)	1.54 (0.85)	2.53 (0.69)
shot_accuracy_pct	29.22 (20.60)	31.91 (13.77)	40.30 (9.65)
crosses_p90	1.22 (1.62)	2.26 (1.75)	1.40 (1.62)
interceptions_padj_p90	0.96 (0.37)	0.74 (0.44)	0.24 (0.17)
tackles_won_padj_p90	1.07 (0.48)	1.16 (0.54)	0.56 (0.34)
fouls_committed_padj_p90	0.87 (0.39)	1.31 (0.55)	1.28 (0.61)
offsides_p90	0.06 (0.07)	0.16 (0.17)	0.53 (0.33)

gradient from defenders through midfielders to forwards rather than separated groups, and that gradient is the honest description of the geometry. Figure S14 shows the distributions and the three embeddings and Figure S18 the principal component structure, while Figures S15 to S17 give the archetypes this sample returns.

S11.3 The same pipeline across the big five

Repeating the reduced-sample pipeline unchanged on Spain, Germany, Italy and France for the earlier of the two seasons returns the same answer everywhere. The declared rule selects 2 clusters in all 5 leagues. The silhouette at that choice spans 0.255 to 0.286 across the five, bootstrap stability spans 0.907 to 0.946, and agreement with listed position groups spans 0.190 to 0.270. Each of those three ranges is narrow, and the third is uniformly weak: in no league does the partition recover the positions the source already records. The elasticity result generalises as well, since all 15 defensive elasticities fitted independently inside each league lie between 0.081 and 0.692, every one below the value of one that the conventional possession adjustment assumes. Figure S22 shows the comparison.

S11.4 What a supervised model sees

Predicting the listed position group from the same 18 features, gradient boosting reaches an accuracy of 0.803 and a macro F1 of 0.769, and logistic regression reaches 0.811 and 0.786, against a majority class baseline of 0.494. The features clearly do carry position, so the weak agreement between clusters and positions reflects the clustering finding a different and equally real axis rather than an empty feature set.

The attribution carries a finding that follows directly from the withdrawal. With possession and passing volume gone, positions are identified largely by absence: defenders by low shooting and low key passes, forwards by low interceptions, and midfielders by an intermediate profile on both. A classifier of this kind trained before January 2026 would have identified a defender by what he did with the ball, and it now identifies him by what he does not do. Players the classifier places in the wrong group overlap with the players the clustering flagged as hybrids, at 0.556 for gradient boosting against an enrichment of 1.32 over independence, which is corroboration rather than a strong result because the hybrid flag is itself broad.

The finishing above role regression illustrates a hazard worth recording. Predicting non-penalty

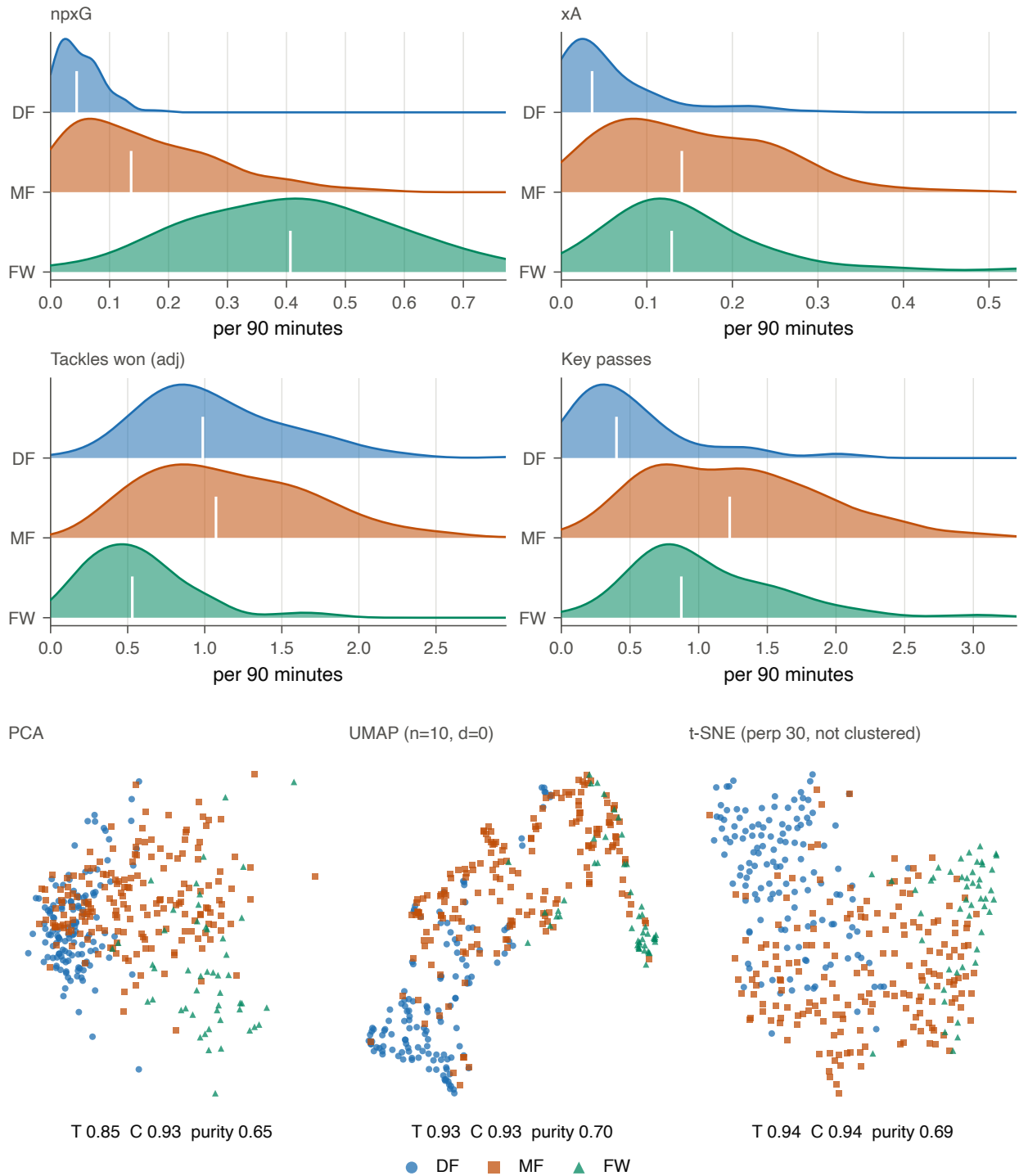


Figure S14: Reduced sample. Top, distributions of four representative per-90 statistics across position groups, where separation is clear for shot volume and defensive set actions and much weaker for creation. Bottom, two-dimensional embeddings of the same feature set coloured by listed position group; the defender to forward gradient is continuous in all three, and t-SNE is shown for visual contrast only and is never used for clustering.

expected goals per ninety from non-shooting behaviour returns a cross validated coefficient of determination of 0.531. An earlier specification that also included xGChain returned 0.892, which looks far better and is in fact circular, since Understat credits xGChain for possessions the player finished himself, so the difference between xGChain and xGBuildup contains the target and correlates with it at 0.904. Even at the honest value roughly half the variance is unexplained, so the residual leaderboard of Figure S35 is an illustration rather than a measurement of finishing ability. Figure S23 shows the confusion and the per-class importance, and Figure S24 the full attribution.

S11.5 The third dimension and team signatures

Two components are a convenient way to draw the space and a poor way to describe it. On the reduced sample the first two carry shooting volume and wide creation and together account for 0.521 of the variance, while the third carries duels and discipline, loading on fouls committed, cards and tackles won, and is orthogonal to both of the others, so a planar projection does not compress it, it deletes it. Retaining it raises captured variance to 0.634. Adding the dimension does not change the headline geometry, since the defender to forward gradient remains continuous rather than resolving into islands.

The third dimension does change the club picture. Club centroids correlate with final league position at -0.861 on the first component and -0.695 on the second, but only 0.209 on the third. Shooting volume and wide creation both track league position; the physical dimension of duels, fouls and cards essentially does not. A club does not finish higher by fouling more or less, which needs stating because the duelling axis is a full component of player variation carrying almost no information about team success. The three-dimensional view also settles a structural question by measurement rather than assertion: drawing each archetype centroid as a direction from the origin, the two centroids within a position group have a cosine of -1 for defenders, -1 for midfielders and -1 for forwards. They are exactly anti-parallel, which is forced by standardising within group before a two-way split, so six archetypes occupy three lines through the origin.

Aggregating players into clubs produces the one result in the reduced sample that is both strong and stable. The minutes-weighted position of a club's outfield players on the first principal component correlates with final league position at a Spearman coefficient of -0.771 in the earlier season and -0.783 in the replication season, close to an exact reproduction. Since the first component is essentially attacking involvement, this says that clubs whose players collectively shoot and create more finish higher, which is unsurprising in itself; the near identical coefficient across two independent seasons shows the team level summary is measuring something stable even where the player level partitions are not. Hierarchical clustering on archetype composition groups the 20 clubs into 3 style families. Figures S25 and S26 show the three-dimensional views, Figure S27 the club signatures, and Figure S28 each club's players against the rest of the league in the shared embedding, on axes shared across all twenty panels.

S11.6 Similarity, replication and sampling noise

Nearest neighbour search within position group recovers football sense for players whose roles are defined by the surviving statistics, and fails informatively for those whose roles are not. Full backs return full backs and penalty box forwards return penalty box forwards. The instructive failures are the midfielders: a holding midfielder's nearest neighbours are creative attacking midfielders, because with no tackles by pitch third, no blocks, no clearances, no pressures and no passing volume, the only things left that distinguish him are his set piece deliveries and key passes. The metric finds the players who share that signature and misses the entire defensive half of his job, which Figure S29

makes visible.

242 players are eligible in both reduced-sample seasons. The declared rule independently selects the same number of clusters in every scope in both seasons, so the structure replicates, while membership replicates only partly: the adjusted Rand index between seasons is 0.598 globally, falling to 0.526 for defenders, 0.381 for midfielders and 0.453 for forwards, with the transitions in Figure S30.

Empirical Bayes shrinkage toward the position group mean, with the degree of shrinkage set by minutes played, gives the sharpest test of whether the partitions describe playing roles or sampling noise. The global partition survives: only 5 of 366 players change cluster, the adjusted Rand index between the original and shrunk partitions is 0.945, and the rate at which players move falls monotonically with minutes played, exactly as the noise account predicts. The within group partitions fare worse and not equally. For defenders the index falls to 0.453, a partition largely dissolved by removing sampling noise, while midfielders retain 0.770 and forwards 0.626. Figure S31 shows the change rate against minutes and Figure S32 the effect on every feature.

Read alongside the bootstrap and replication results, the three tests do not deliver a single uniform verdict on the reduced sample, and reporting them as though they did would overstate the case. Defenders fail all three there, forwards fail two of the three, and midfielders pass all three. The conclusion that survives every test on the reduced sample is narrower than a single summary sentence would suggest: the global division between attacking and defensive involvement is a real property of players, the midfield split is trustworthy, and the defender split should not be used to characterise an individual on that feature set. The primary sample, with twice the features and six times the players, supports all three splits, which is the clearest single illustration of what the richer description buys.

S11.7 Role drift within a season

Every result elsewhere treats a season as one observation per player, which assumes a player's profile is homogeneous across it. Match reports make that assumption testable, and because the withdrawal removed the affected columns from the match schema rather than emptying them, what remains there is fully populated. A full season is 380 reports at seven seconds each, so the scrape is restricted to one club, Tottenham, chosen because its eligible players span all six archetypes and therefore give drift the widest range to move across. 38 matches were retrieved and 26 players accumulated at least one complete rolling window of 6 matches. This is an illustration of a limitation rather than an estimate of its size, and it should be read as one.

The match feature set has 10 of the eighteen features used elsewhere, since the Understat expected goals block has no match equivalent. Rather than assume the coarser space is a fair stand-in, we refit the season clustering using only those features and compare it to the partition reported throughout; the two agree at an adjusted Rand index of 0.903, so drift measured within the reduced space is interpretable.

Most players do not drift. 5 of 26, or 19 percent, change cluster at least once during the season, and the remainder occupy a single cluster throughout. The players who stay put are the ones whose roles are least ambiguous, holding midfielders on one side and the recognised attackers on the other, while the movers are predominantly full backs and wide players whose attacking involvement varies with the fixture. For four fifths of players a season aggregate describes a single role and the homogeneity assumption holds; for the remaining fifth it averages over two. Figure S33 shows the trajectories.

S11.8 Age and archetype

The two archetypes within a position group differ mainly in attacking involvement, and involvement is plausibly something a career passes through rather than a fixed trait, so the association with age is a natural test. It is tested within position group, because archetypes are nested inside positions and a pooled table would largely recover the fact that forwards and defenders have different age profiles.

Only defenders show a relationship. The crossing and chance creating defenders average -1.65 years younger than the low involvement group, a difference the rank test places at 0.015, and the share of the crossing archetype falls monotonically from the youngest band to the oldest. Midfielders show nothing at all, with a gap of 0.12 years at 0.698, and the global partition is unrelated to age at 0.415. Forwards point the same way as defenders but cannot support the claim, since six of the ten cells in their table have expected counts below five.

The defender result also illustrates a point about test choice that is easy to get wrong. A chi-square on the banded table returns 0.194 and would ordinarily be read as no association. That test spends its degrees of freedom on any departure from independence, while the pattern here is monotonic, and a rank correlation against age recovers it at 0.209 with a p-value of 0.015. The banded test is not wrong, it is underpowered against the alternative that actually holds, and reporting only the chi-square would have converted a real finding into a null one. Figure S34 shows the shares.

S11.9 Goalkeepers on the reduced sample

Post-shot expected goals, sweeper actions, distribution and cross handling are all absent from the reduced sample, so only shot stopping and workload survive. That version builds two composites instead of five and substitutes the residual from a regression of saves on shots on target faced for post-shot expected goals. The substitute adjusts for the volume of work a goalkeeper receives but not for its difficulty, and it is reported as the weaker instrument it is. For the primary sample, Figure S36 gives the shot stopping ranking, Figure S37 the correlation structure of the goalkeeper block and Figure S38 the composites that block supports. For the reduced sample, Figure S39 gives the clusters and the two surviving composites, Figures S40 and S41 the shot stopping views that remain, and Figure S42 the correlation structure and composite distributions.

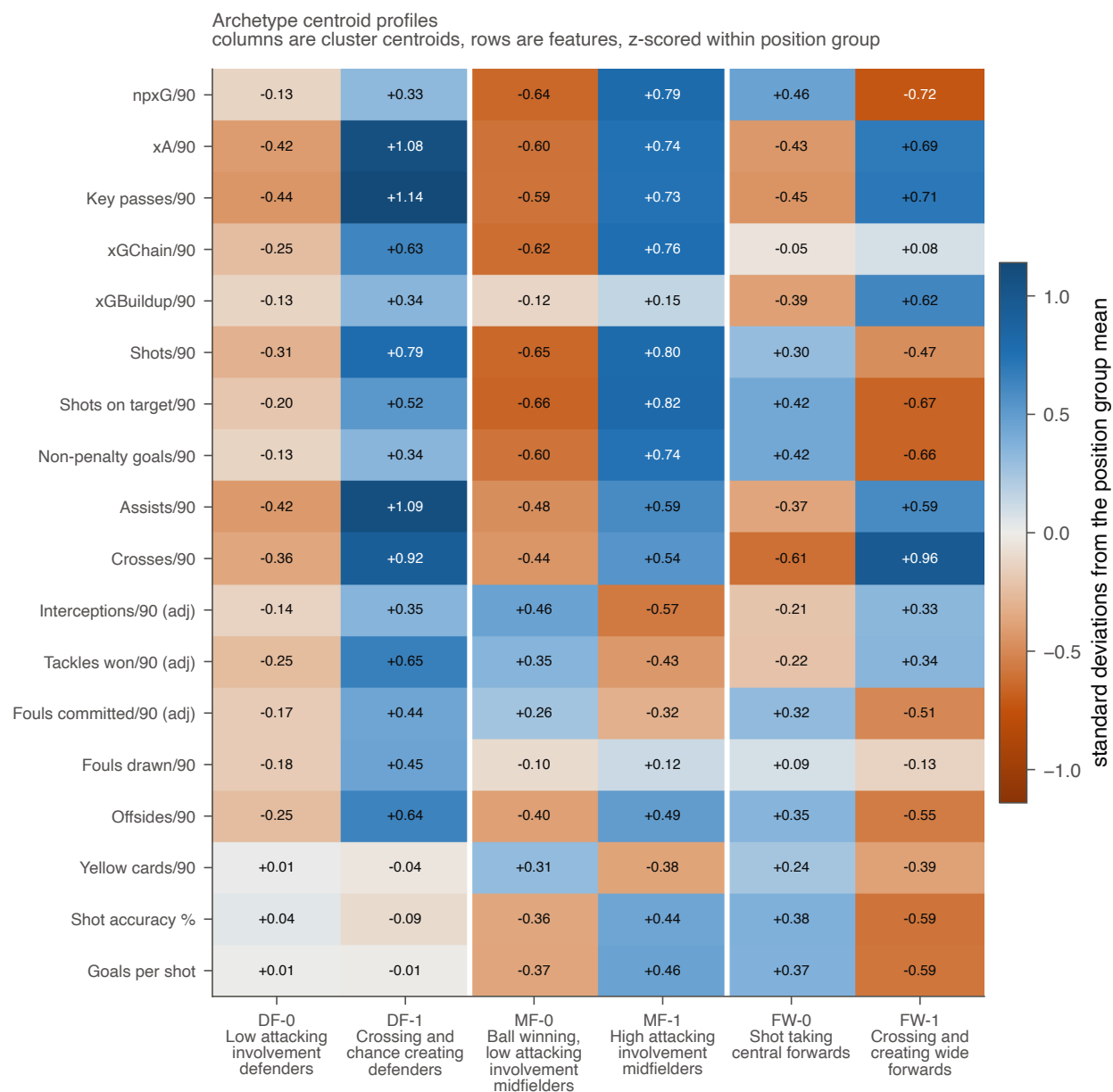


Figure S15: Centroid deviation of each archetype from its position group mean, reduced sample.

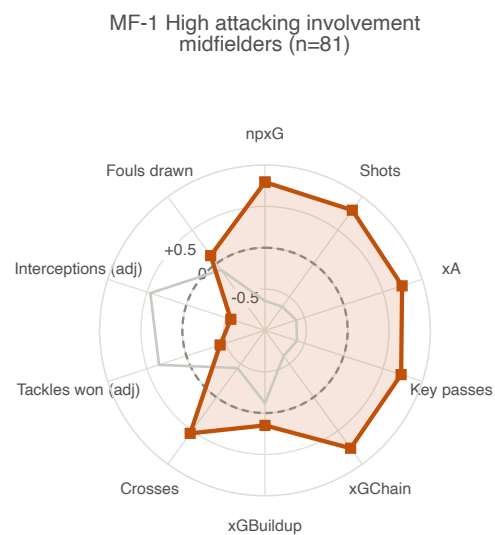
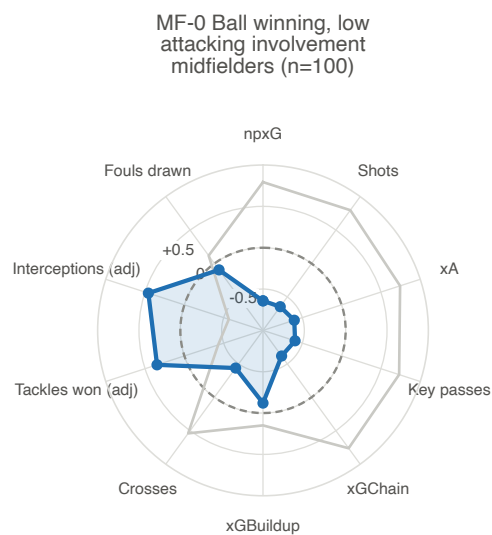
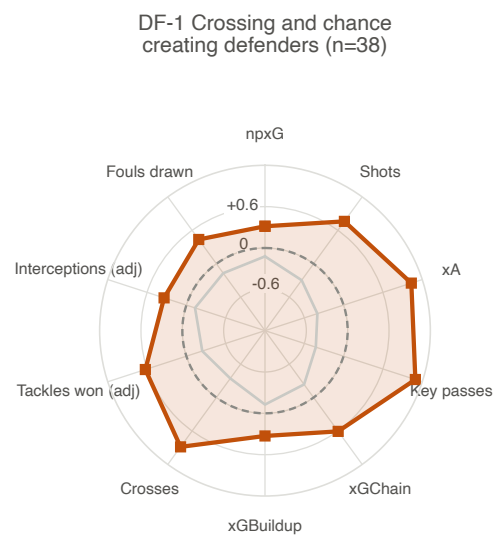
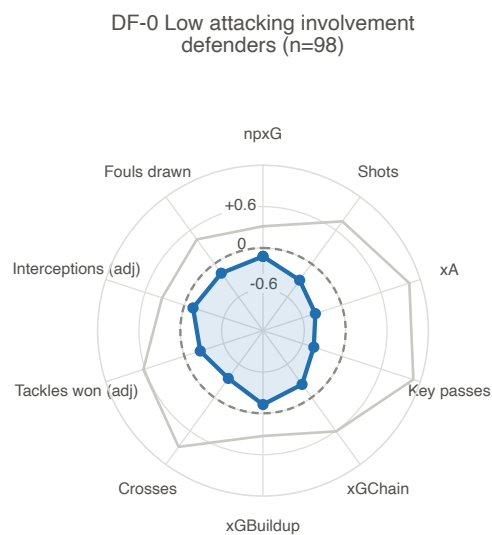


Figure S16: Archetype radars on the reduced sample. Top, defenders. Bottom, midfielders. The two sets of radars in this document are drawn on different feature counts, so their axes are not directly comparable and the useful comparison is of shape rather than extent.

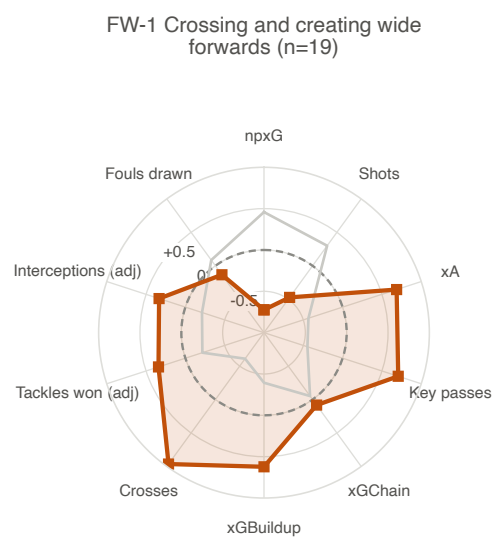
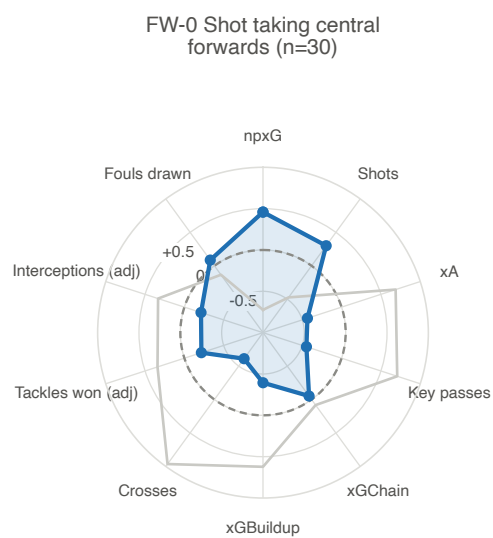


Figure S17: Archetype radars for forwards on the reduced sample, on the same axes as Figure S16.

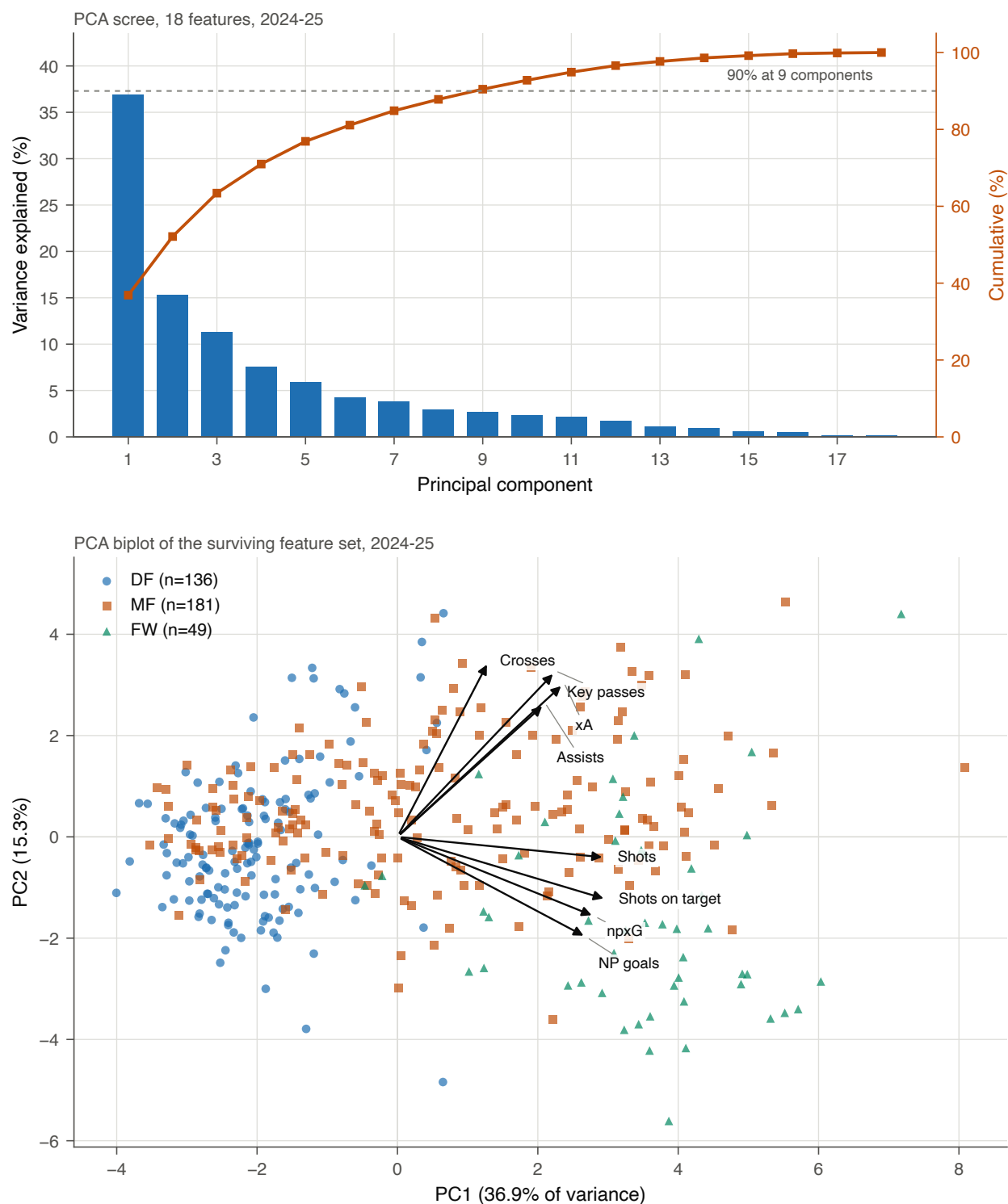


Figure S18: Principal component structure of the reduced outfield feature set. 9 components are required to reach ninety percent of the variance, so the feature set is not close to low rank, and the first three components together account for 0.634. Top, scree and cumulative variance. Bottom, the loadings biplot.

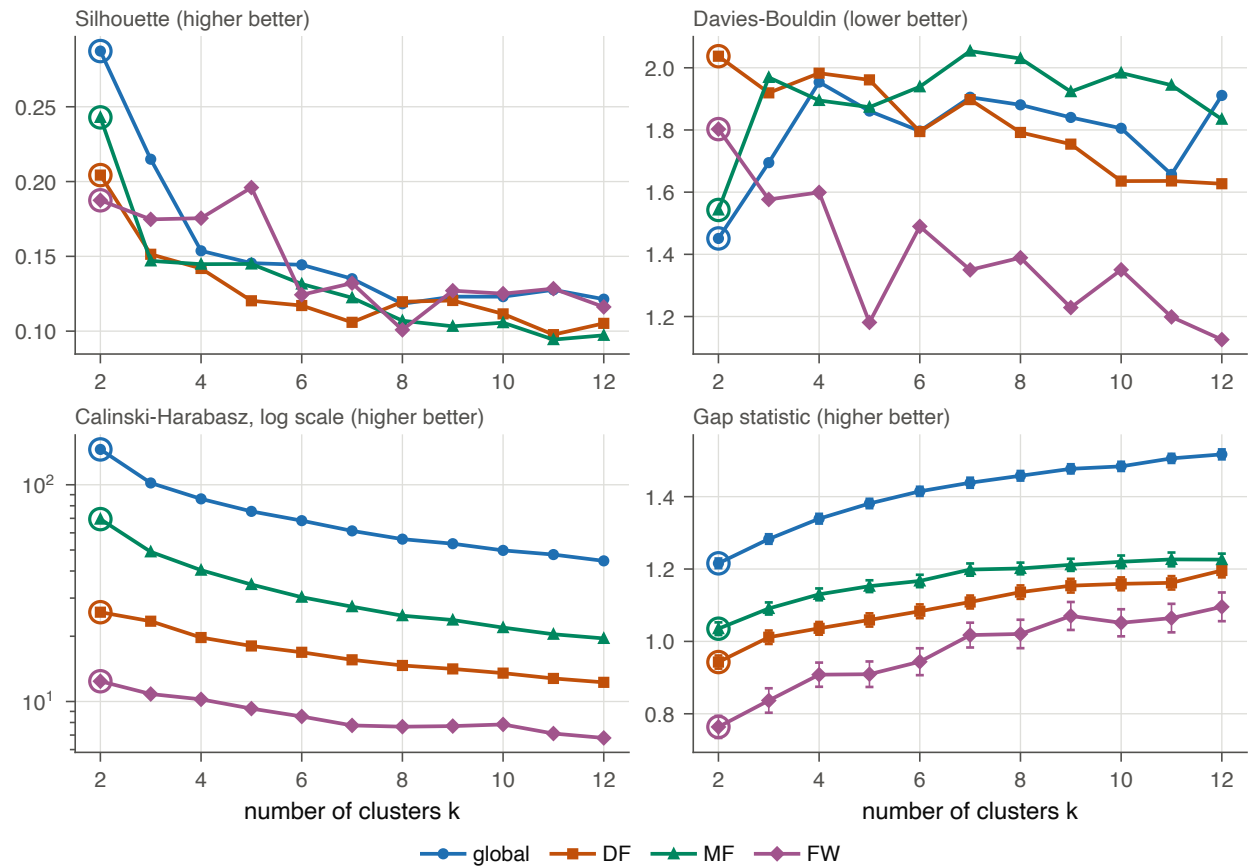


Figure S19: Cluster selection criteria across candidate cluster counts, reduced sample. The consensus rule was fixed before fitting and selects the same value in every scope.

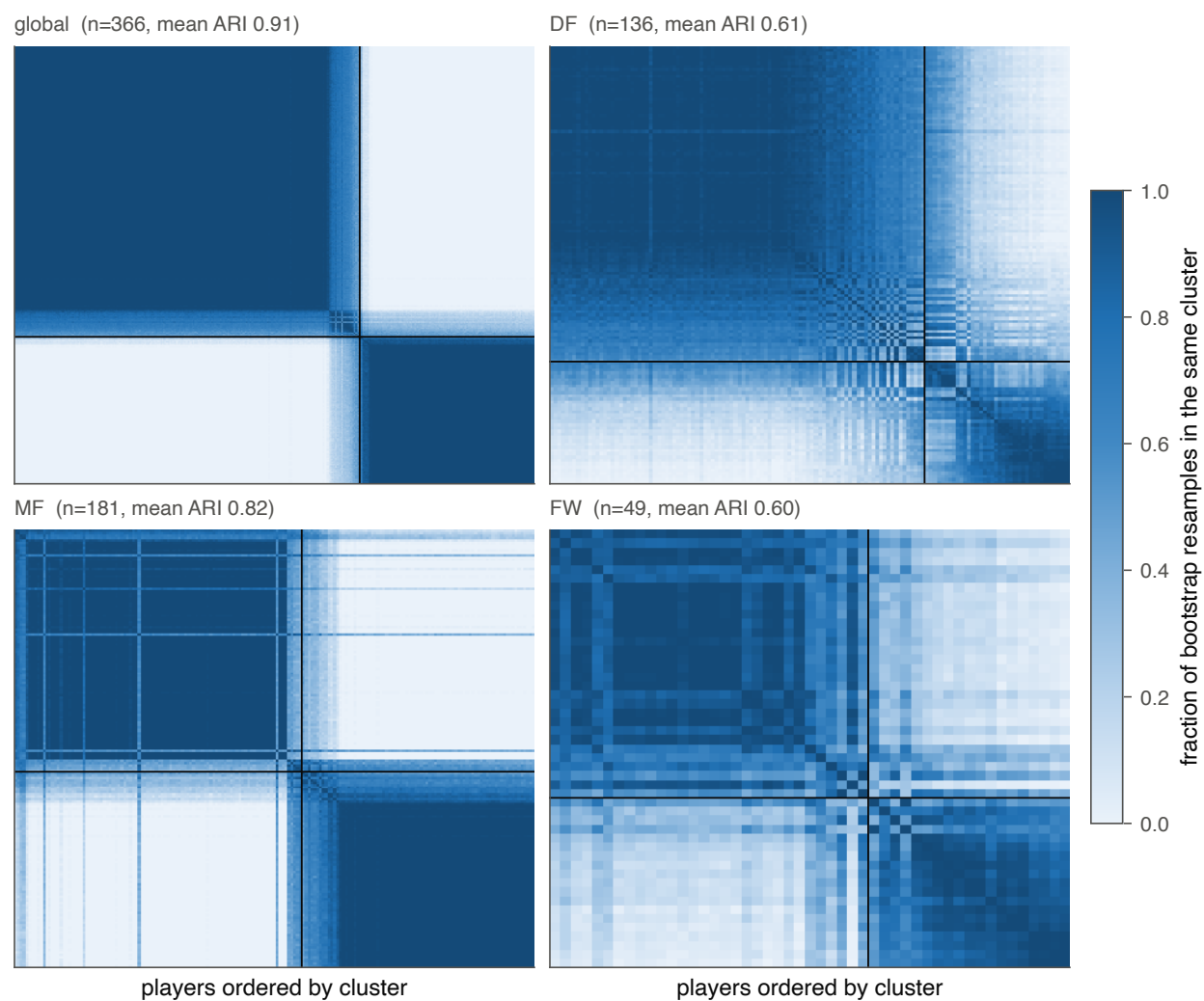


Figure S20: Bootstrap co-assignment structure on the reduced sample, one panel per scope, with players ordered by cluster.

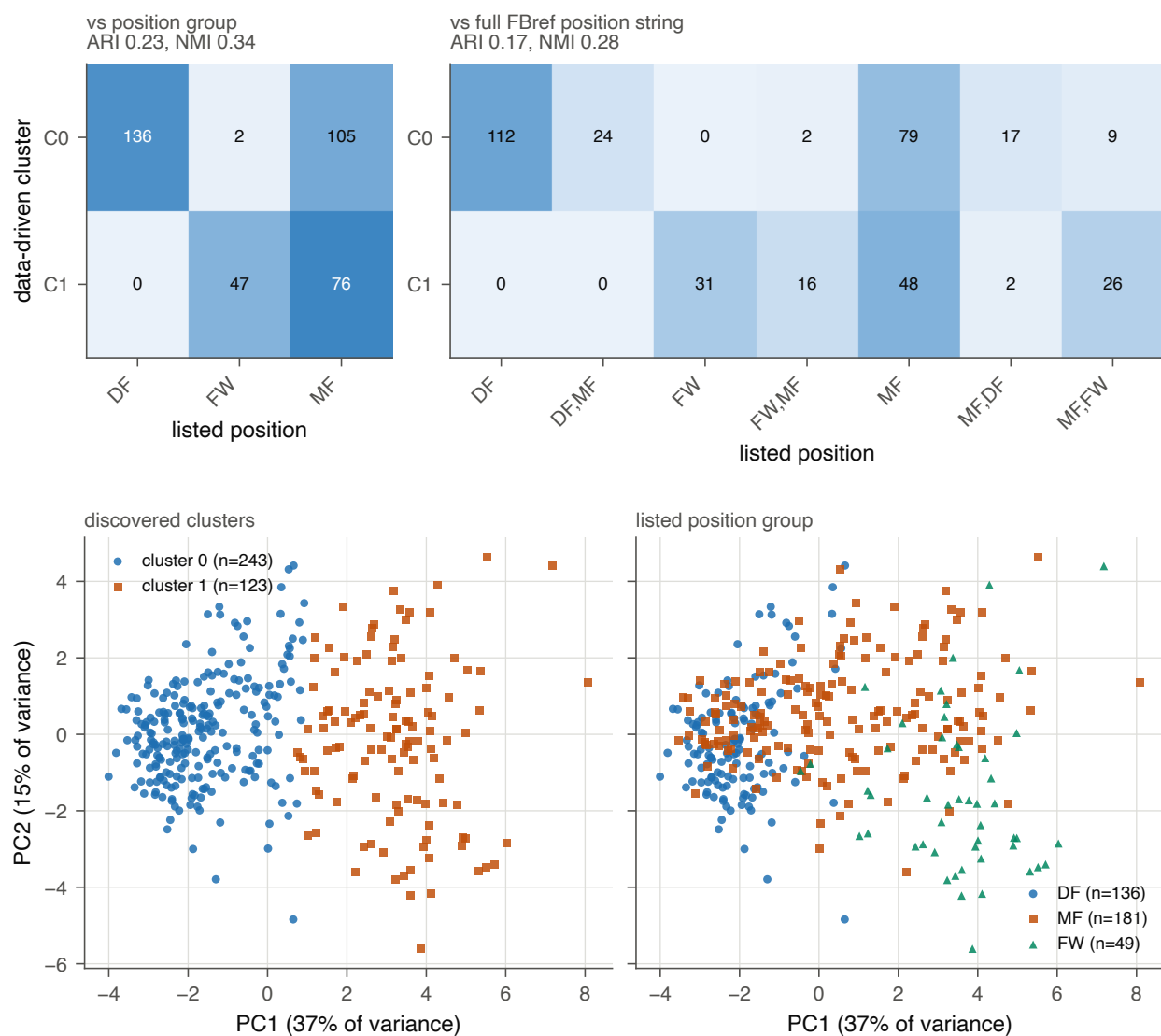


Figure S21: Reduced sample. Top, the selected partition against listed positions. Bottom, the same partition shown in the principal component space.

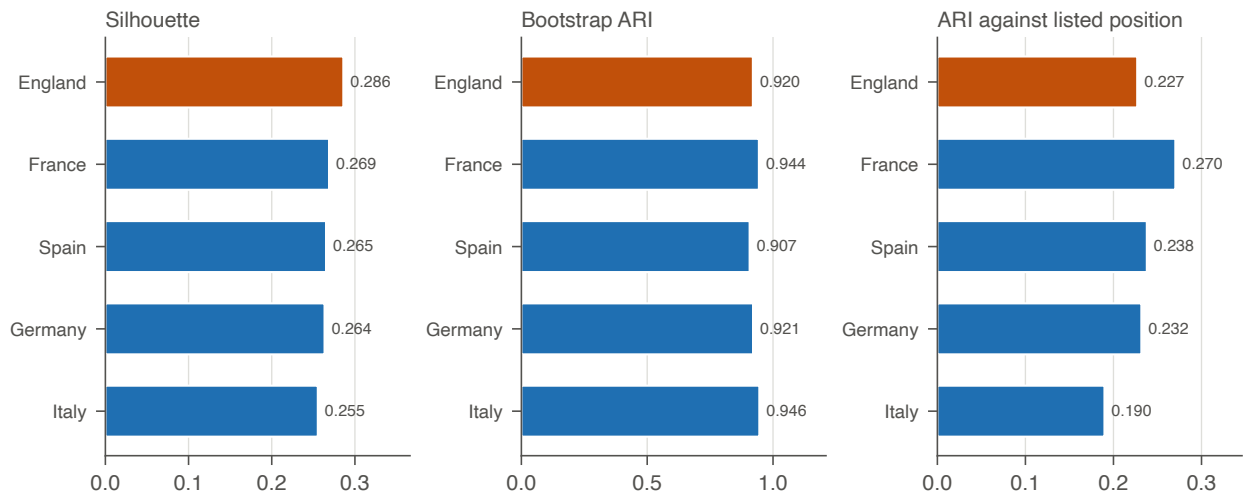


Figure S22: The reduced-sample pipeline applied unchanged to each of the big five leagues, with England, the competition the reduced sample is otherwise drawn from, in the accent colour. The silhouette at the selected number of clusters, bootstrap stability and agreement with listed positions are all close across competitions, and the declared rule selects the same number of clusters in every one.

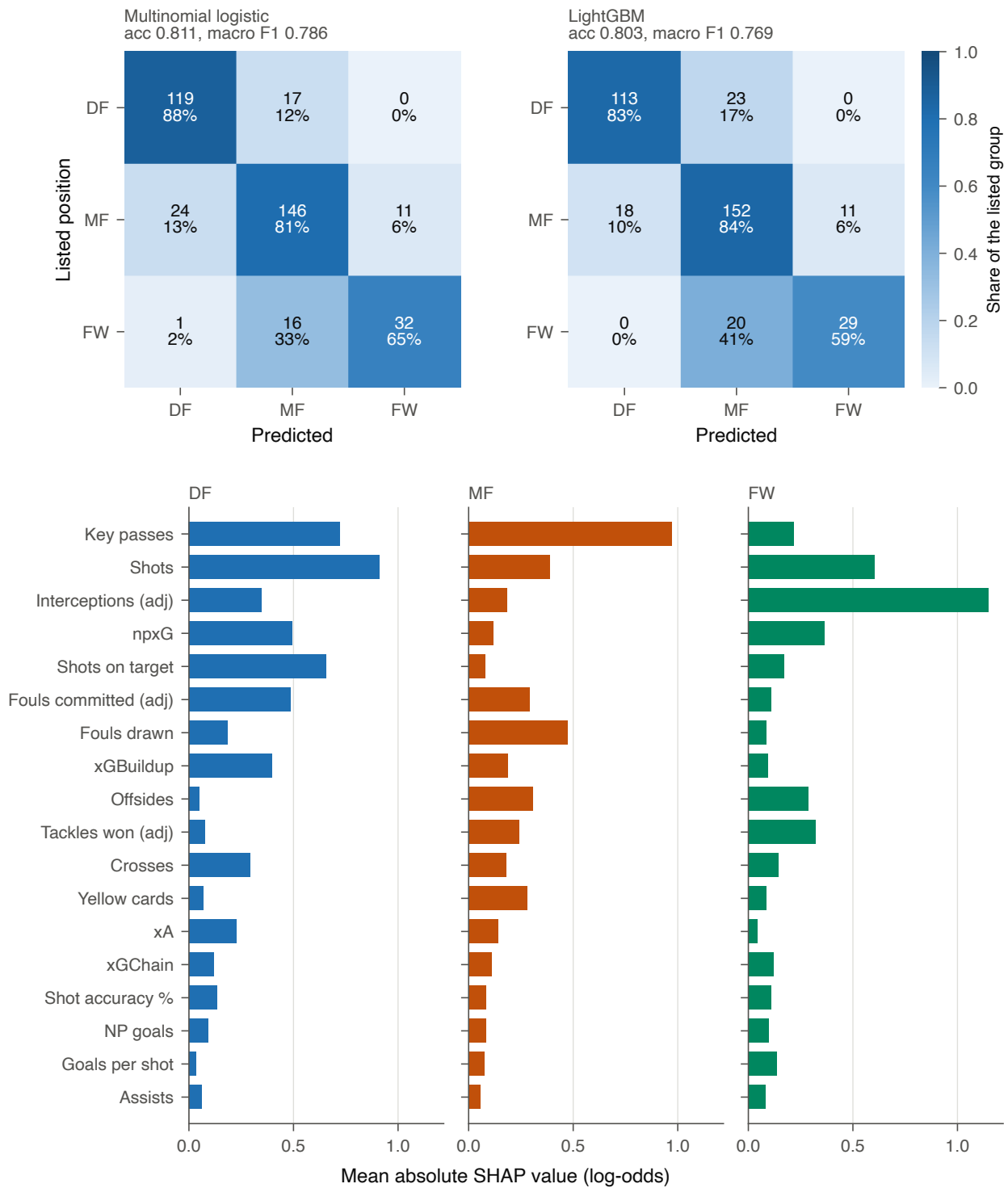


Figure S23: Supervised validation of the surviving feature set, reduced sample. Top, confusion between listed positions. Bottom, mean feature importance per position.

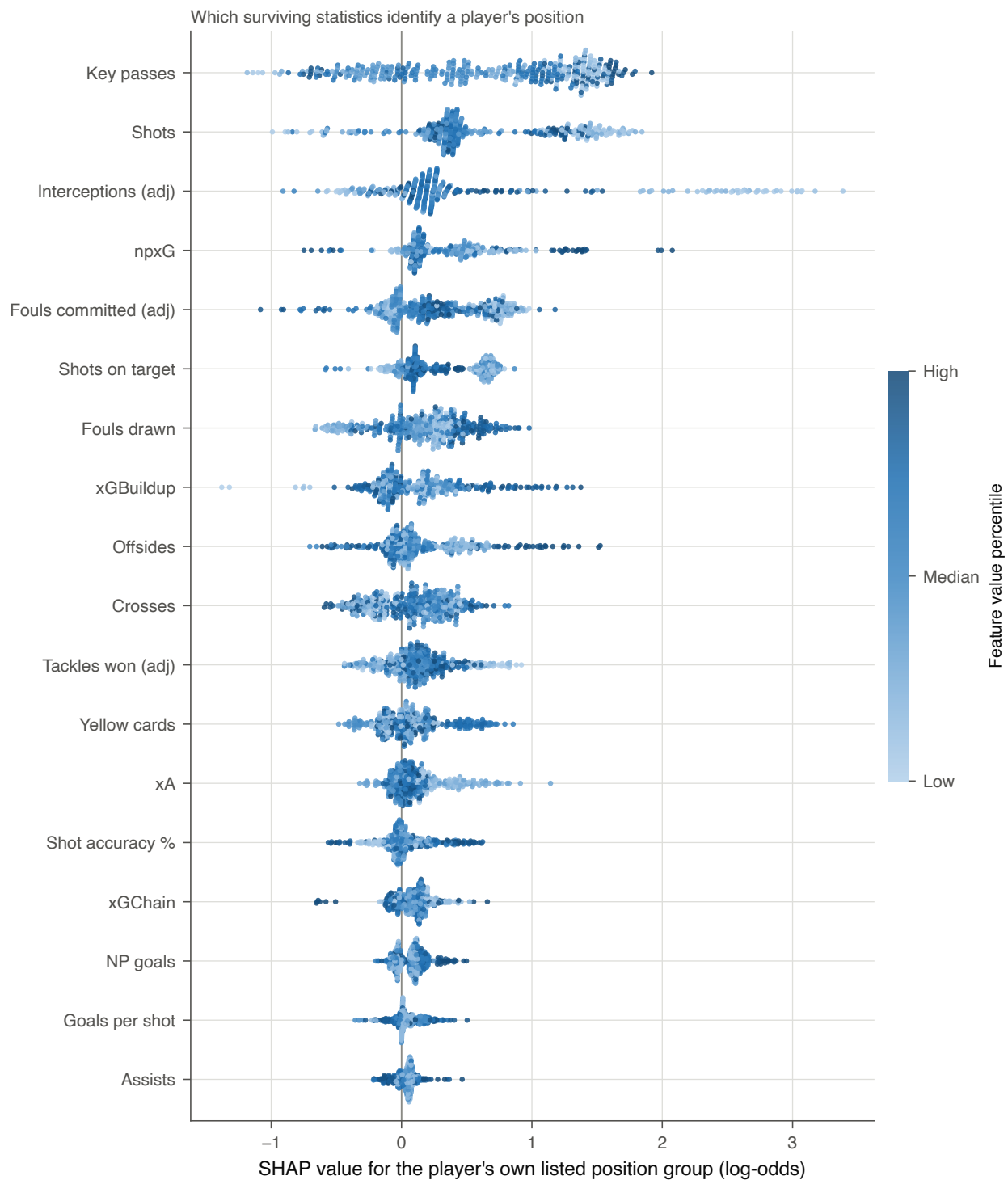


Figure S24: Global SHAP values for the gradient boosted position classifier on the reduced sample. Shooting volume and interceptions carry most of the signal, so a player's position is identified largely by the actions he does not perform.

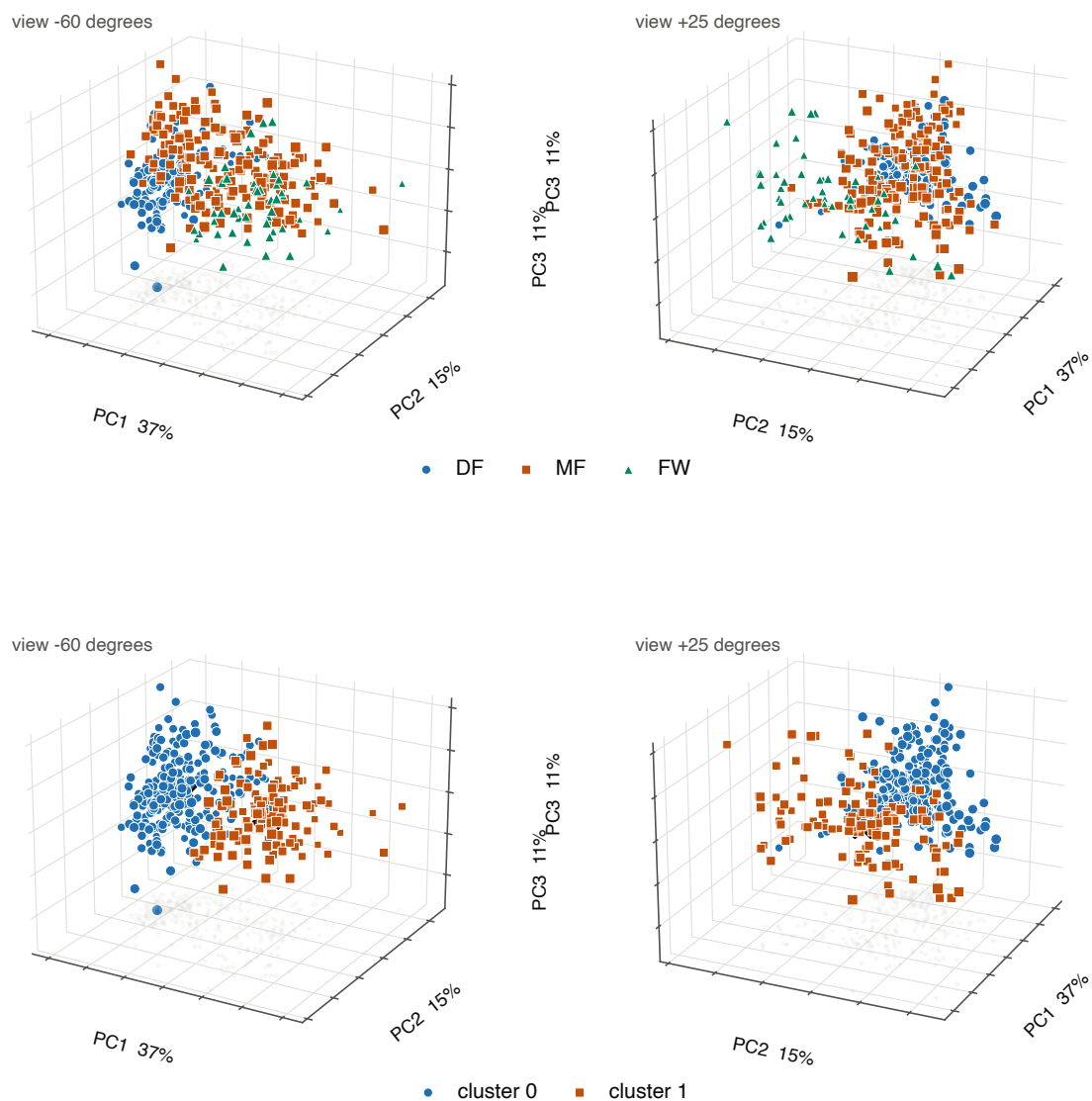
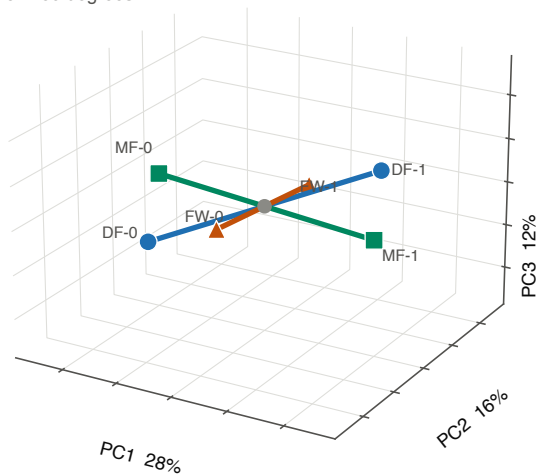
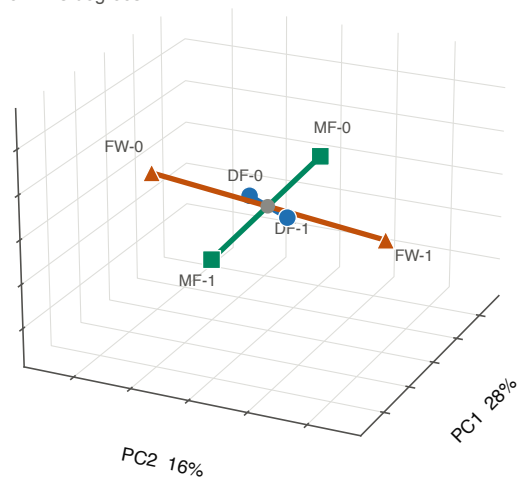


Figure S25: The reduced-sample outfield space in three principal components, each drawn from two viewpoints so that structure can be told apart from the accident of a single projection. Top, players by listed position group, with grey points on the floor supplying a depth cue. Bottom, the selected global partition with centroids marked by crosses.

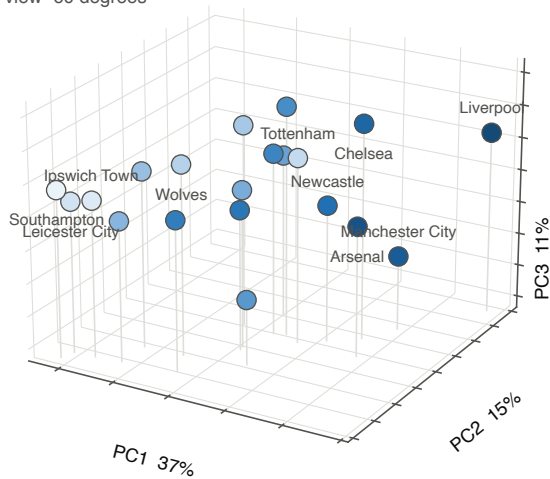
view -60 degrees



view +25 degrees



view -60 degrees



view +25 degrees

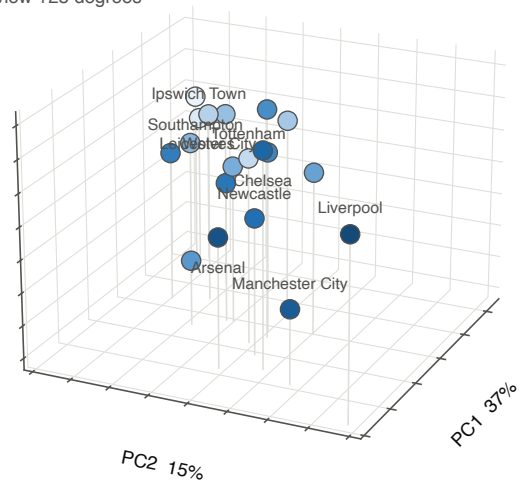


Figure S26: Top, archetype centroid directions drawn as unit vectors, so the figure shows orientation rather than magnitude; each position group forms a single line through the origin, so six archetypes describe three axes. Bottom, clubs as minutes-weighted centroids in three components, shaded by final league position with darker shading for a higher finish.

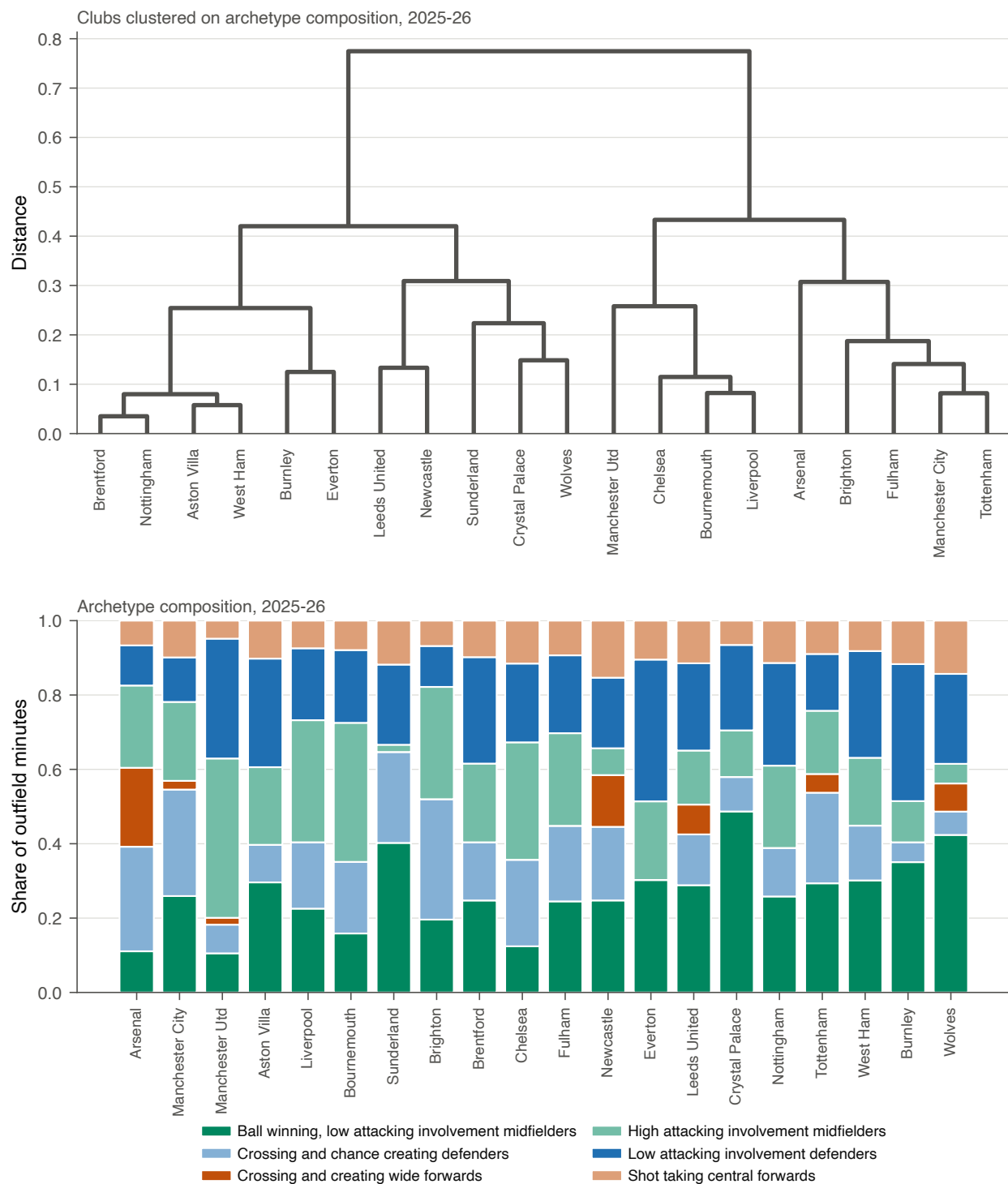


Figure S27: Club signatures on the reduced sample. Top, style families from hierarchical clustering on archetype composition. Bottom, archetype composition by club, ordered by final league position.

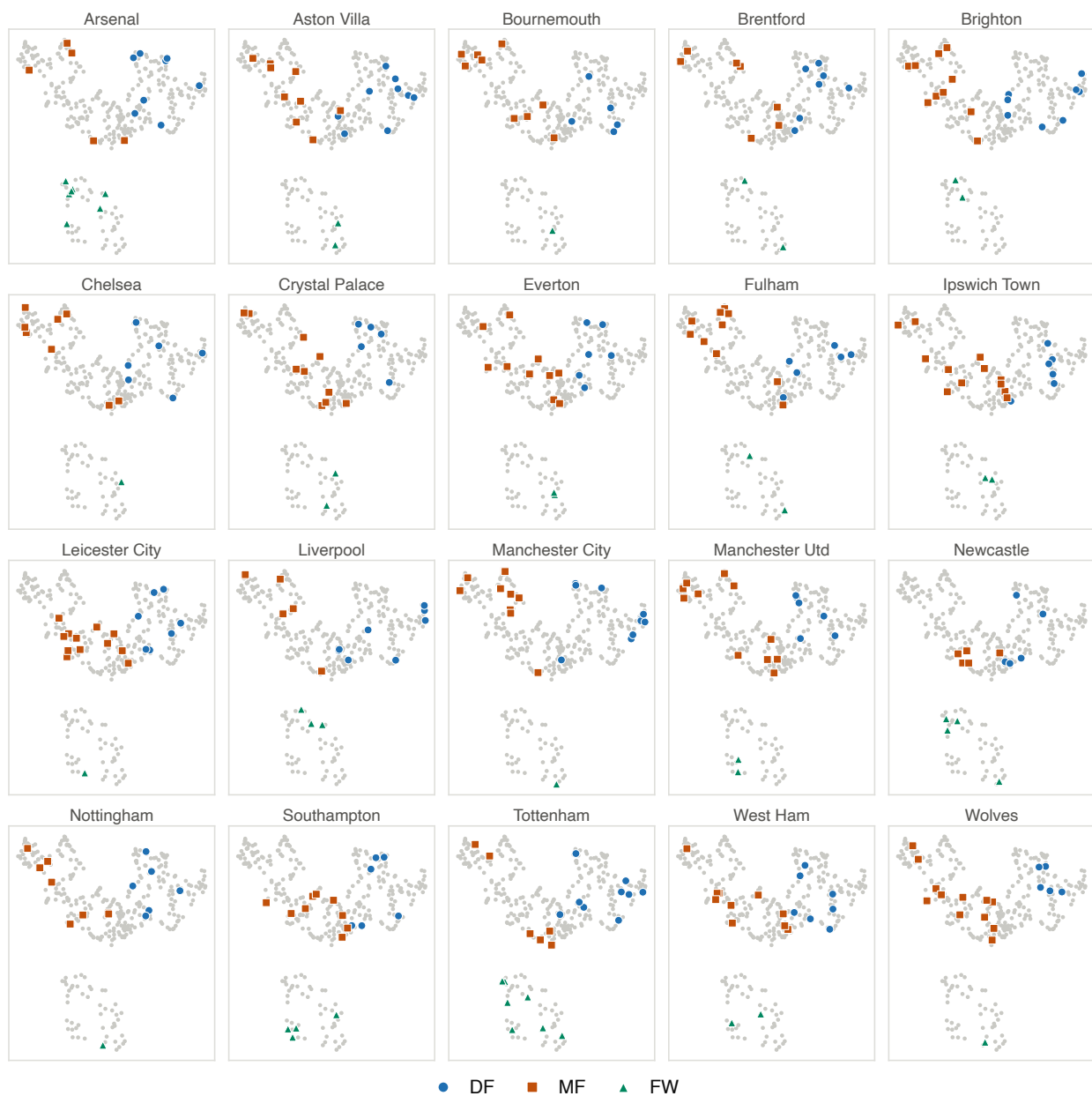


Figure S28: Each club's outfield players highlighted in the shared embedding, with the rest of the league in grey. Axes are shared across all twenty panels, so clubs can be compared directly, and they differ visibly in how far their defenders sit toward the midfield region.

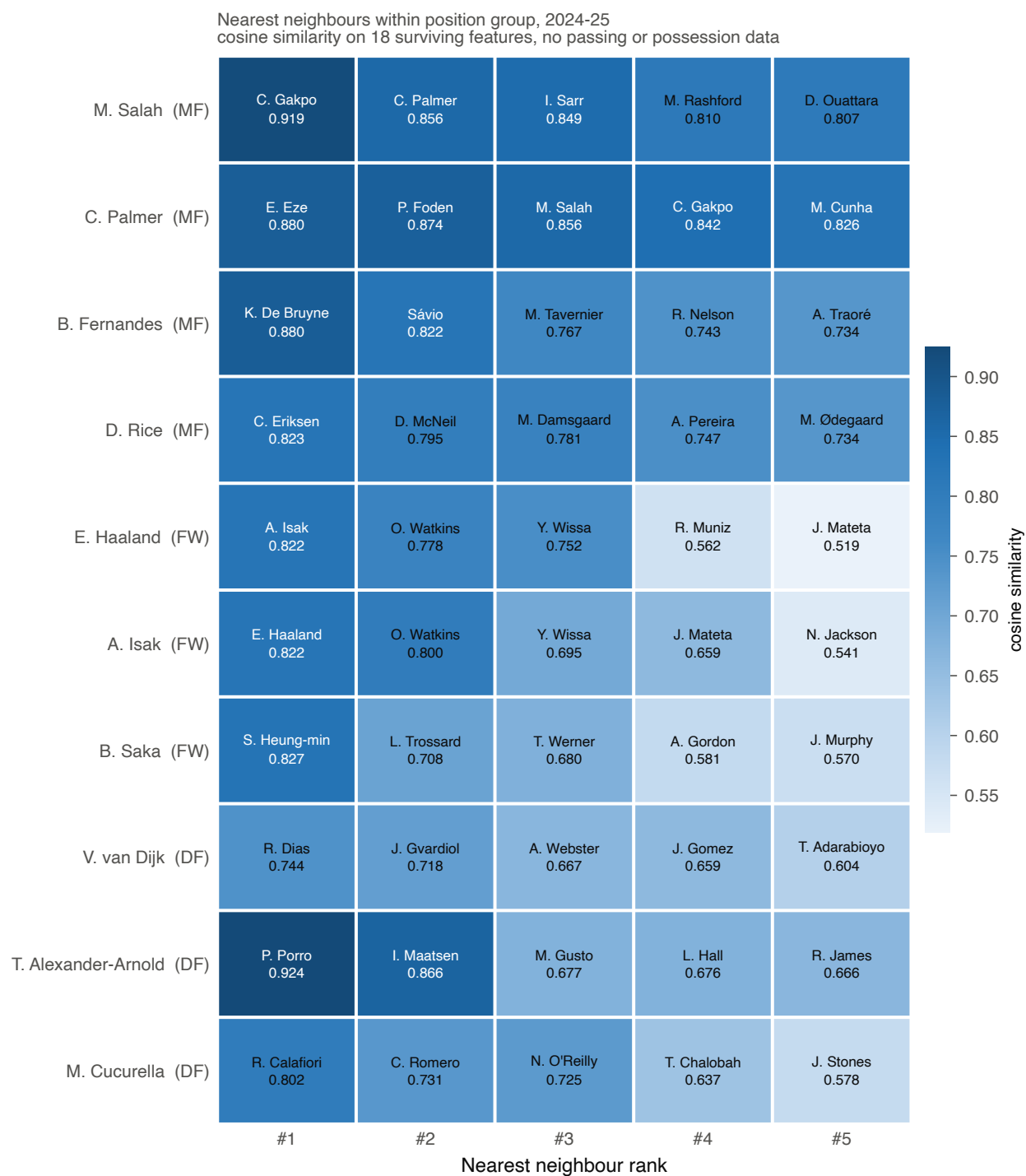


Figure S29: Nearest neighbour similarity structure within position group, reduced sample.

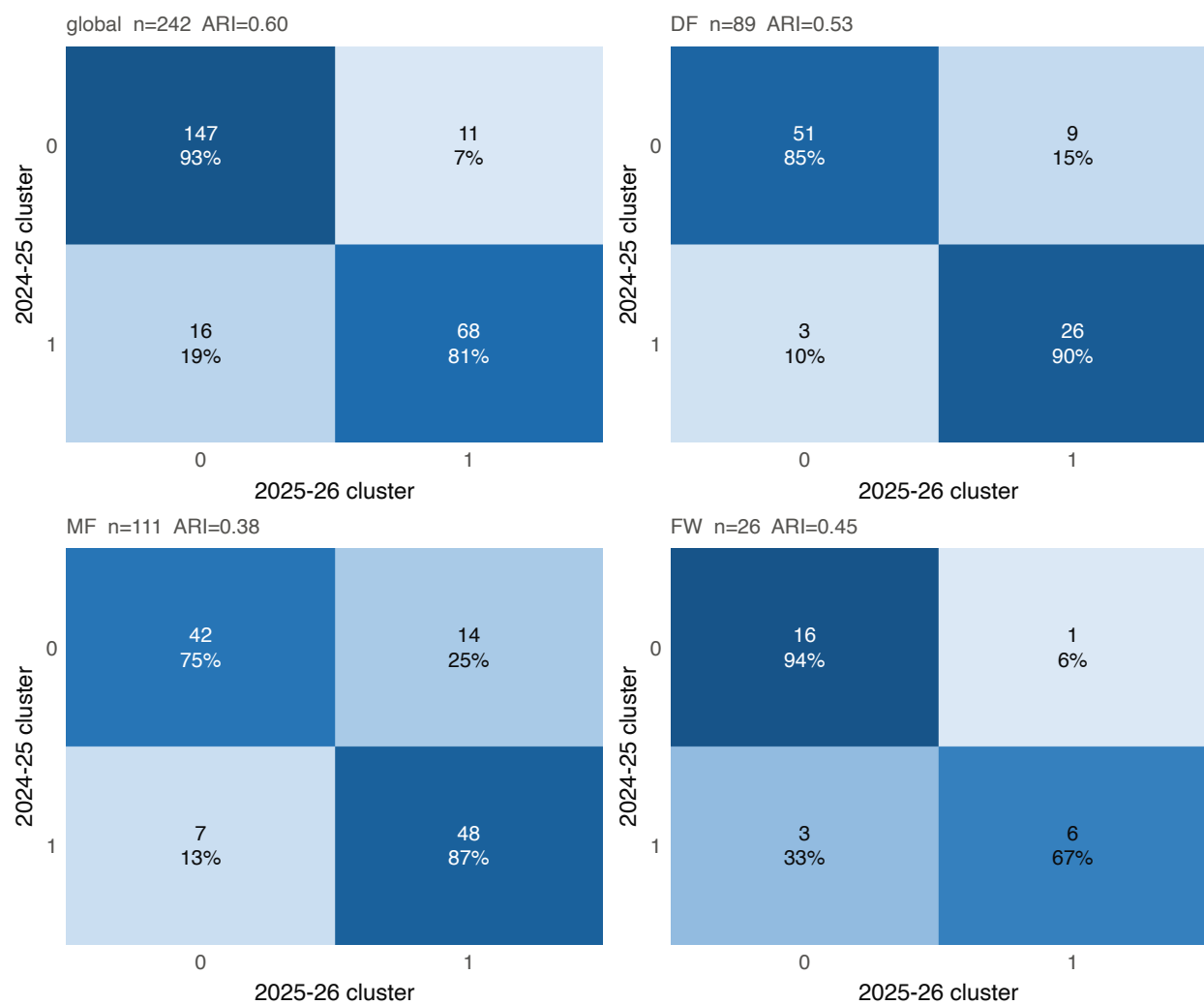


Figure S30: Between-season cluster transitions on the reduced sample, for the 242 players eligible in both seasons.

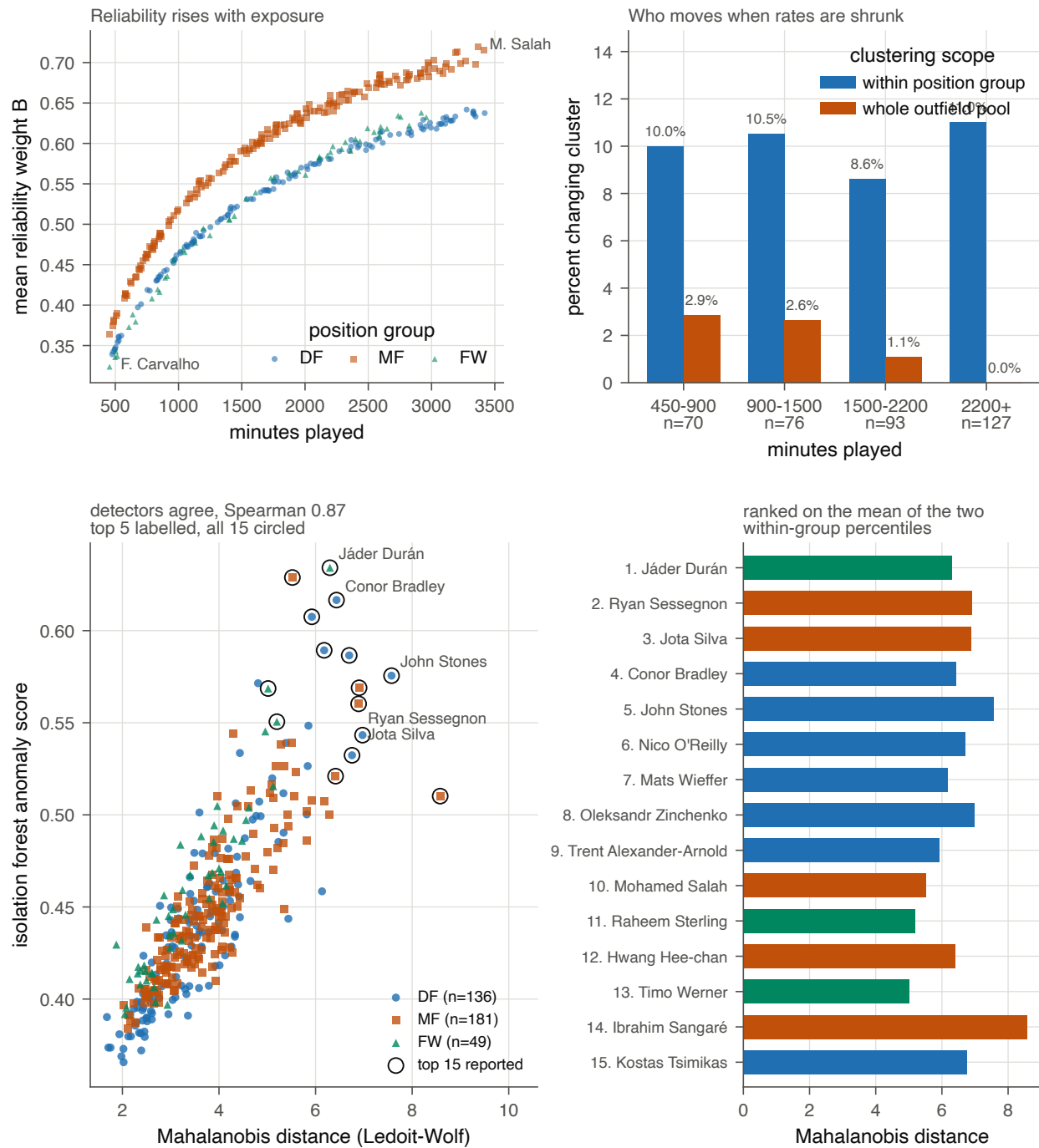


Figure S31: Reduced sample. Top, reliability and cluster change rate against minutes played, which falls monotonically as the noise account predicts. Bottom, the two outlier detectors against each other within position group.

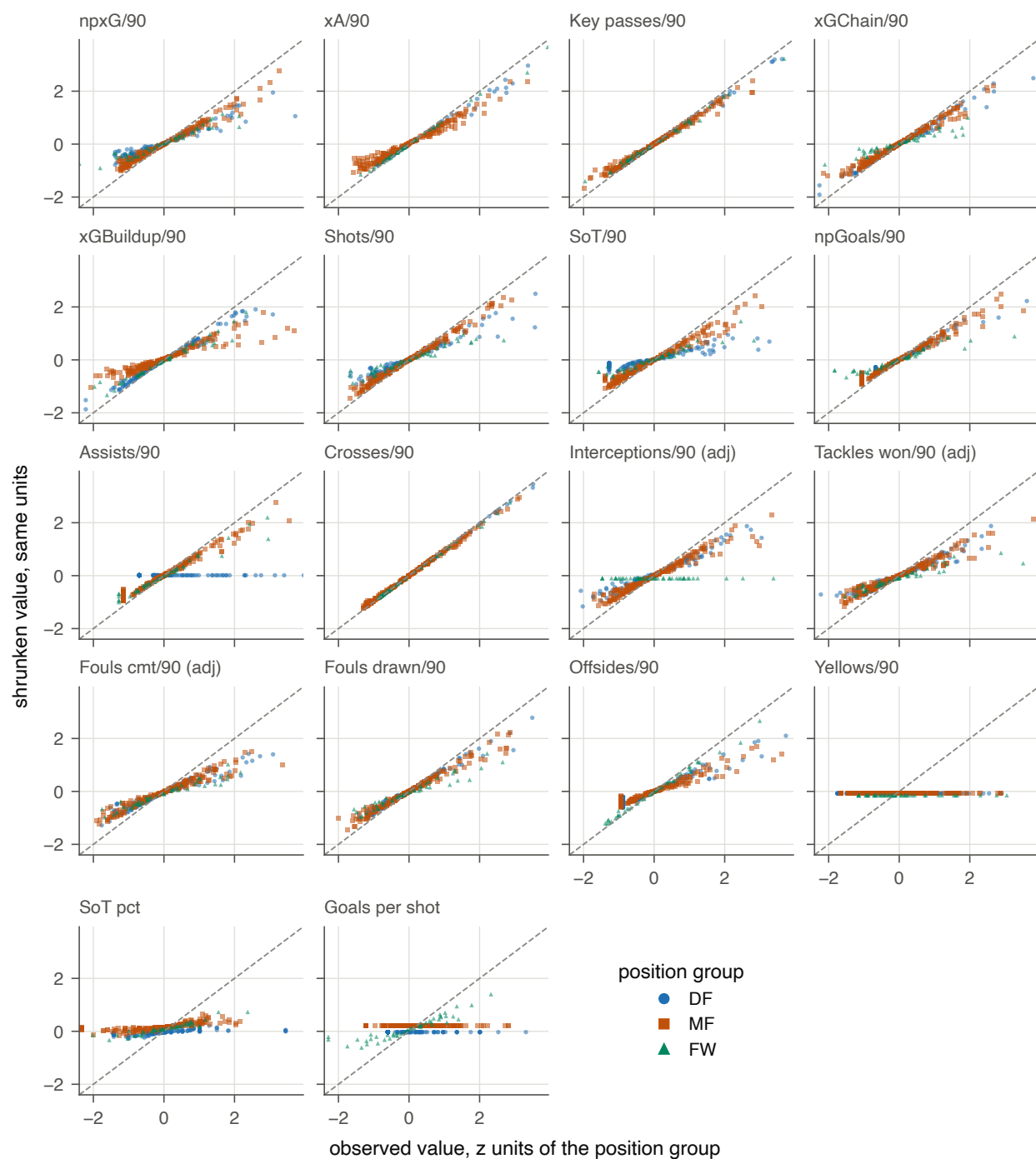


Figure S32: The effect of empirical Bayes shrinkage on every feature of the reduced sample, with the shrunk value against the observed one, one panel per feature.

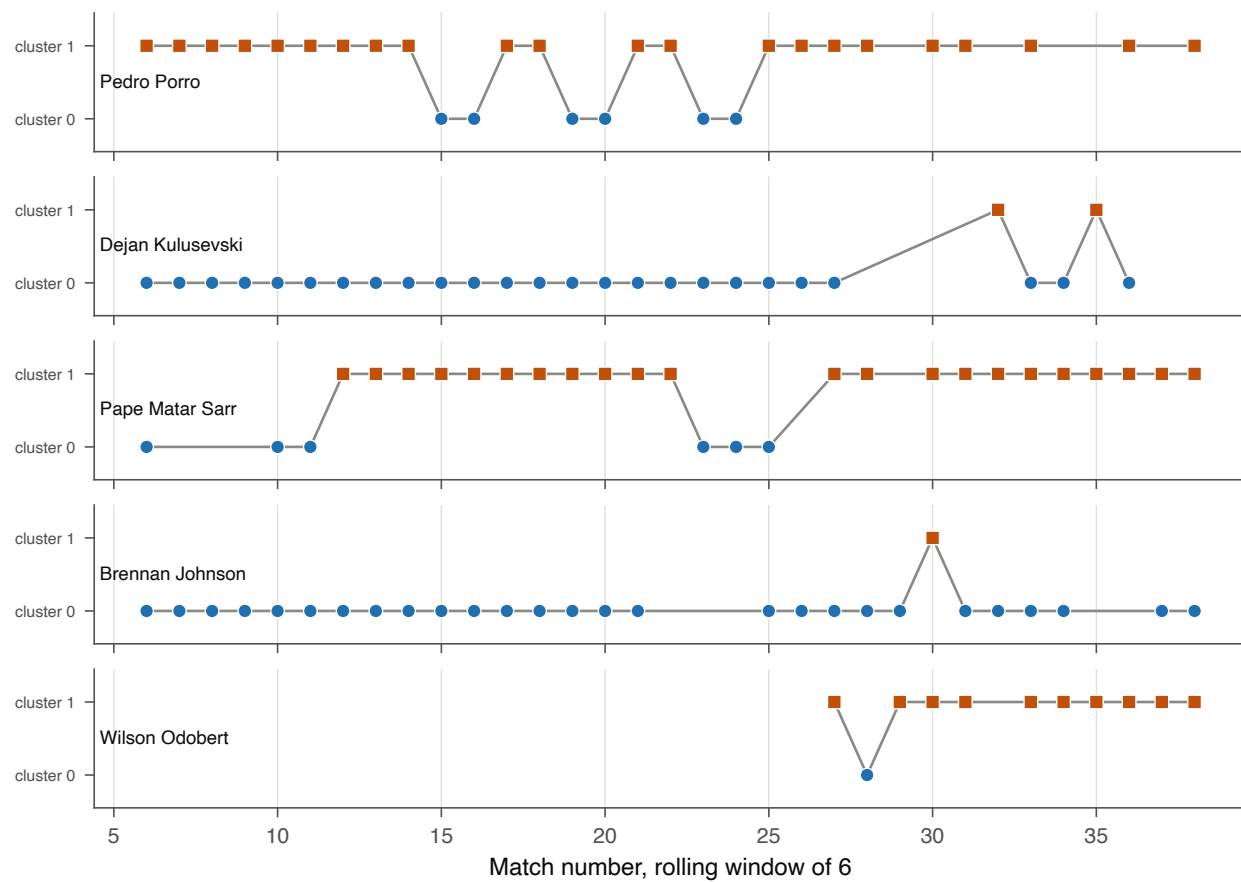


Figure S33: Rolling cluster assignment across the season for the Tottenham players who change cluster at least once, on a 6 match window. Players not shown occupied one cluster for the whole season.

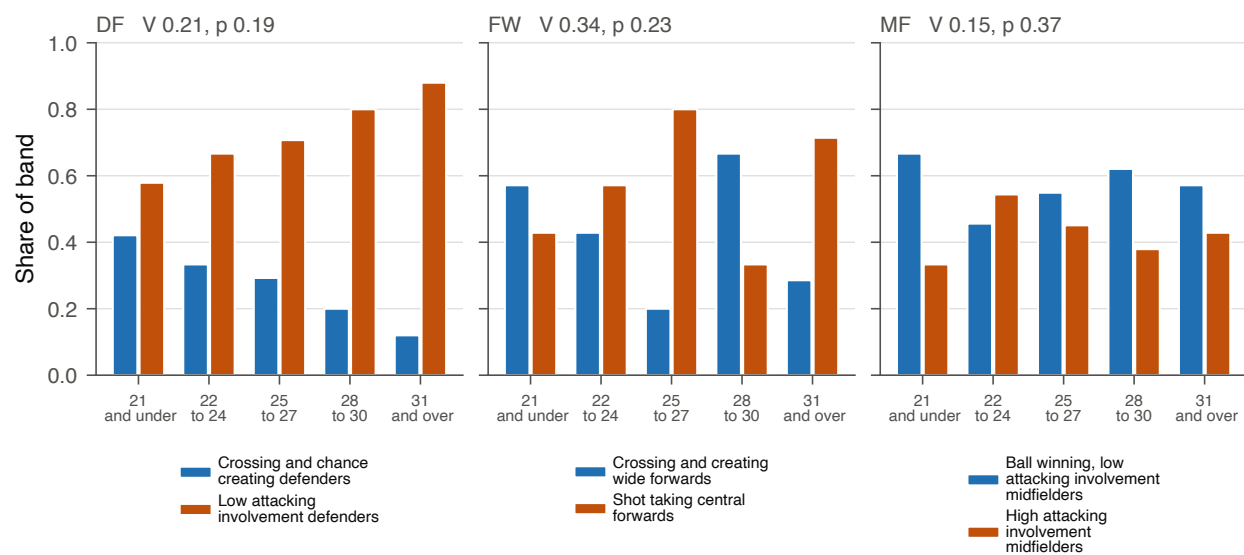


Figure S34: Share of each age band taken by each archetype within position group, reduced sample. The defender panel falls monotonically with age, which a chi-square on bands does not detect and a rank correlation does.

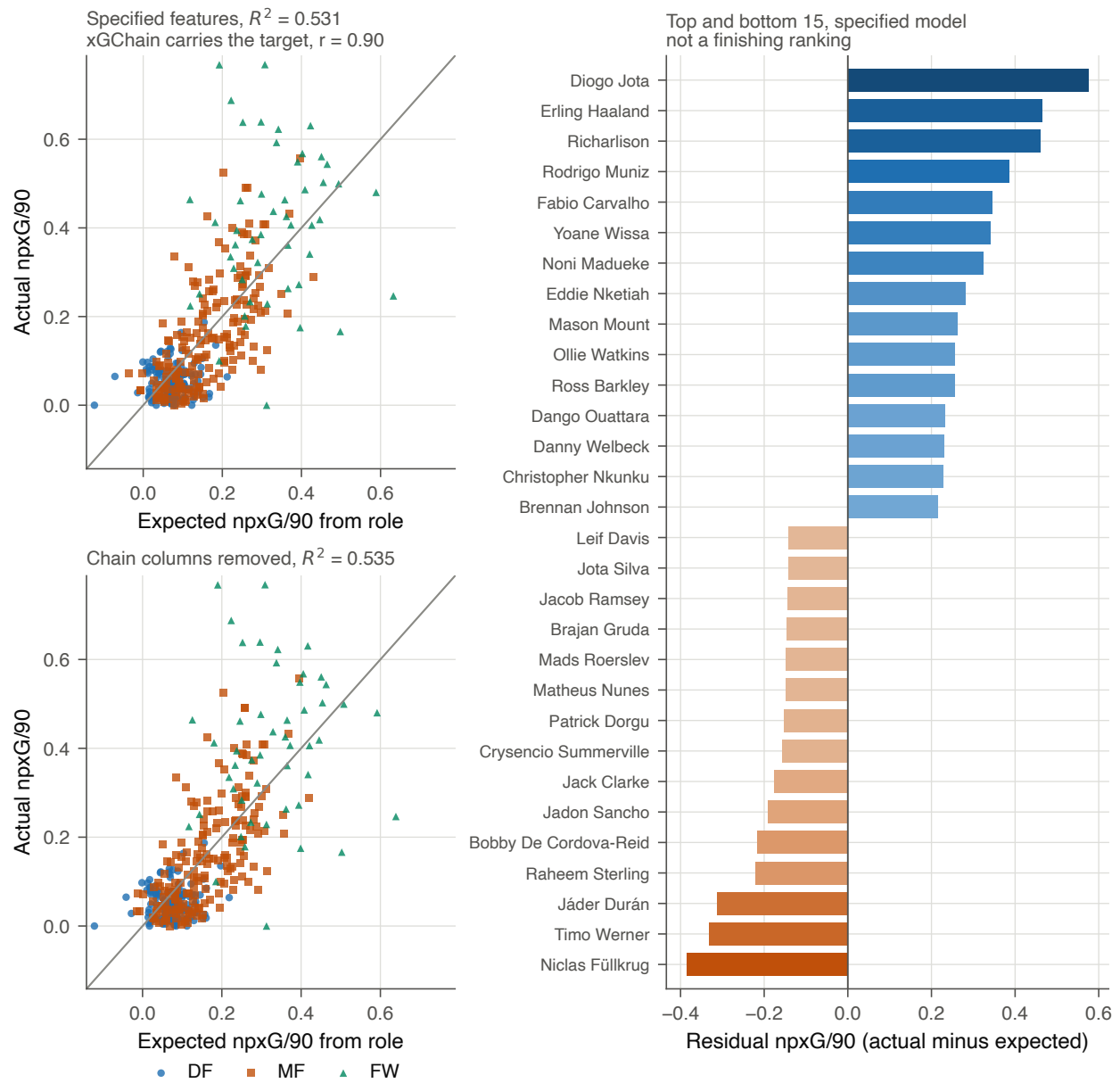


Figure S35: Residuals from the role model on the reduced sample, which should be read as an illustration rather than as a measurement of finishing ability, for the reasons given in the text.

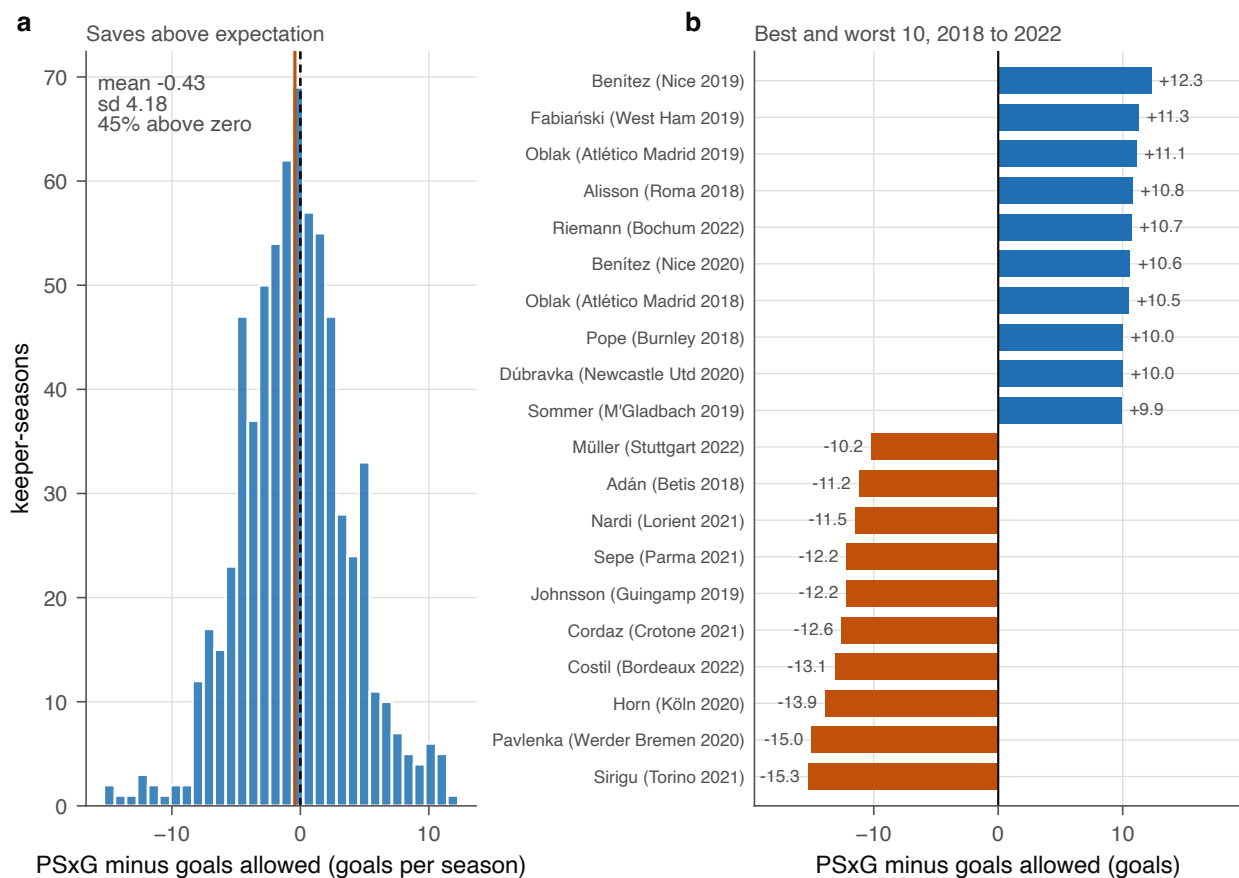


Figure S36: Saves above expectation on the primary sample, as post-shot expected goals minus goals allowed, which conditions on the difficulty of the shots faced rather than only their volume. **a**, the distribution over keeper-seasons. **b**, the best and worst keeper-seasons in the sample.

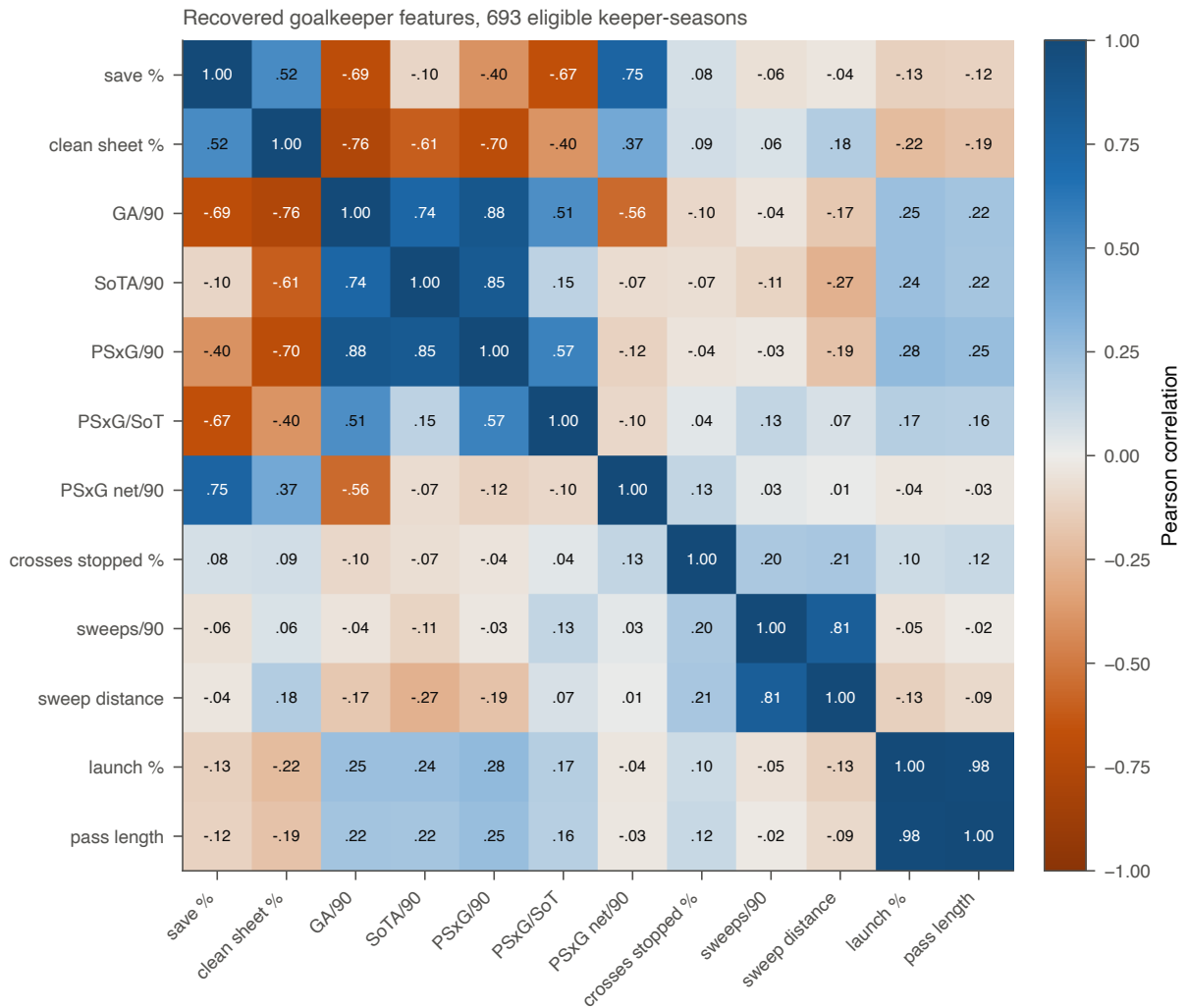
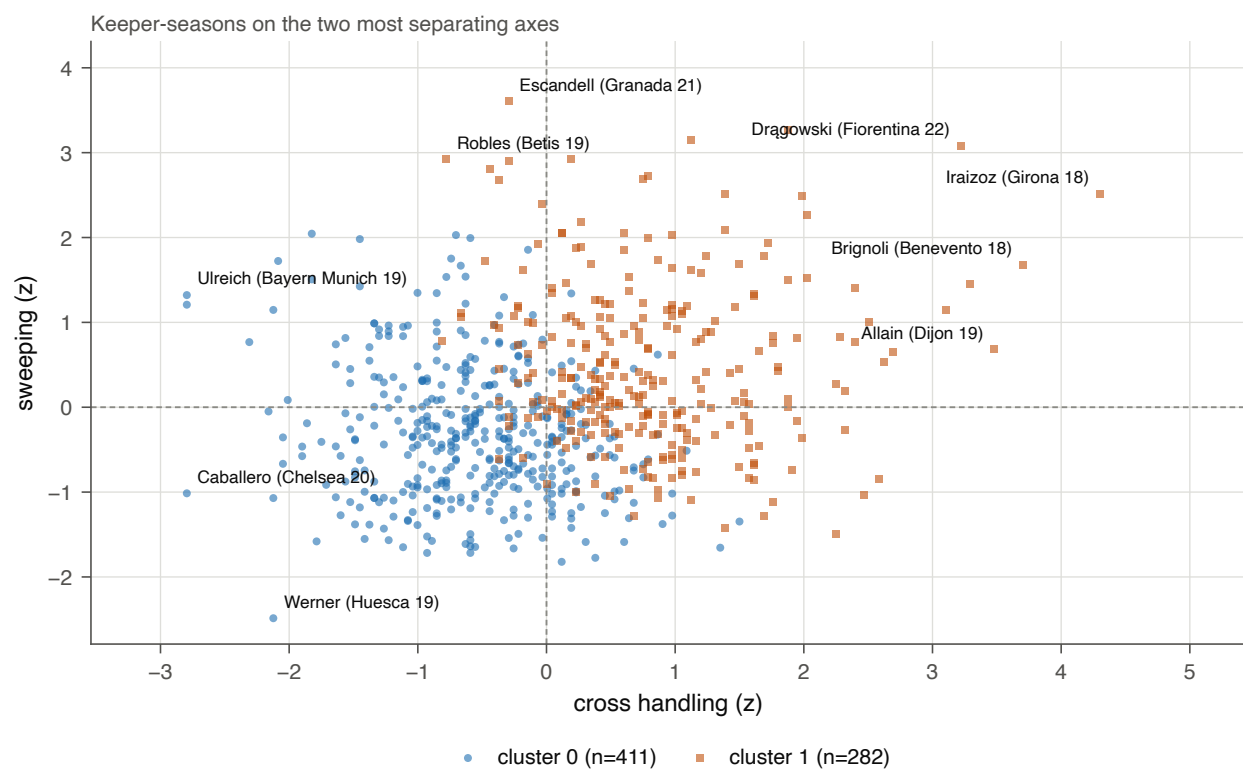


Figure S37: Correlation structure of the 13 goalkeeper features on the primary sample, which is what allows the composite axes the reduced sample cannot build.



eta2 against the partition: cross handling 0.46, sweeping 0.17, workload 0.03.
 Axes chosen for separation, not for football expectation. The most extreme keeper-seasons are named.

Figure S38: Distribution of the goalkeeper composite axes on the primary sample.

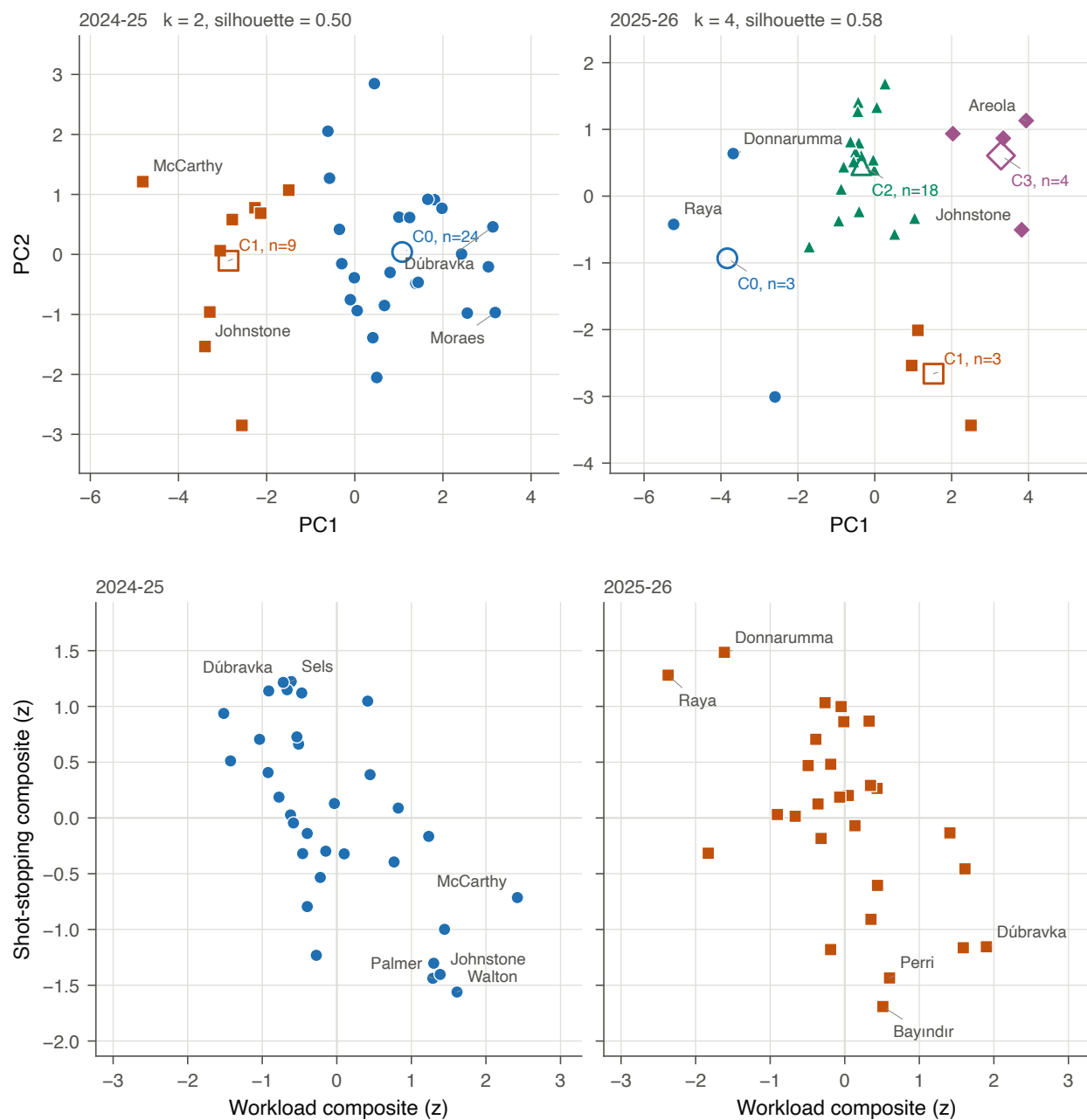


Figure S39: Goalkeepers on the reduced sample. Top, clusters in principal component space. Bottom, the two surviving composite axes.

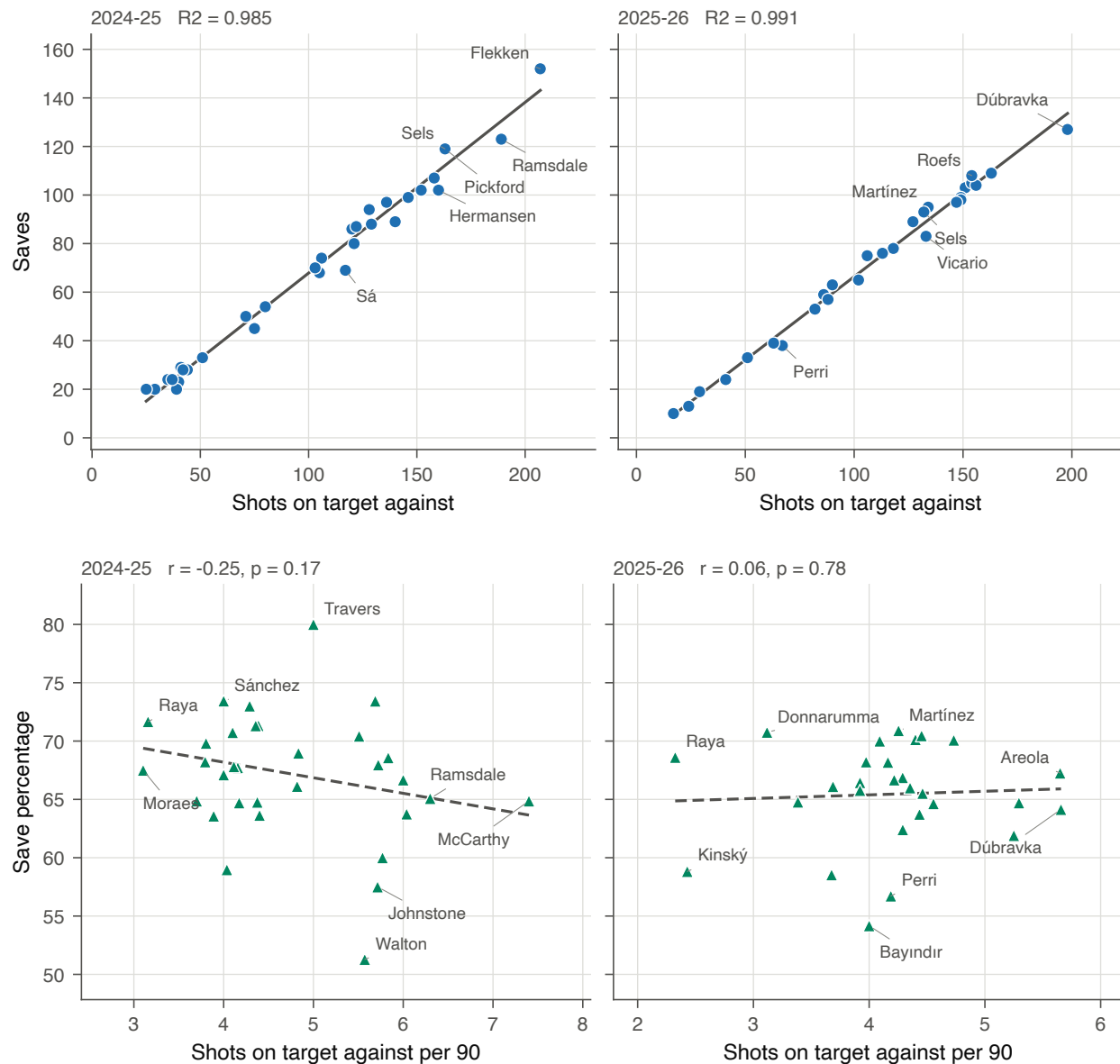


Figure S40: Shot stopping on the reduced sample. Top, saves against shots on target faced. Bottom, save percentage against workload. With expected goals on target withdrawn these are the only shot stopping views the reduced sample supports, and neither separates difficulty from quantity.

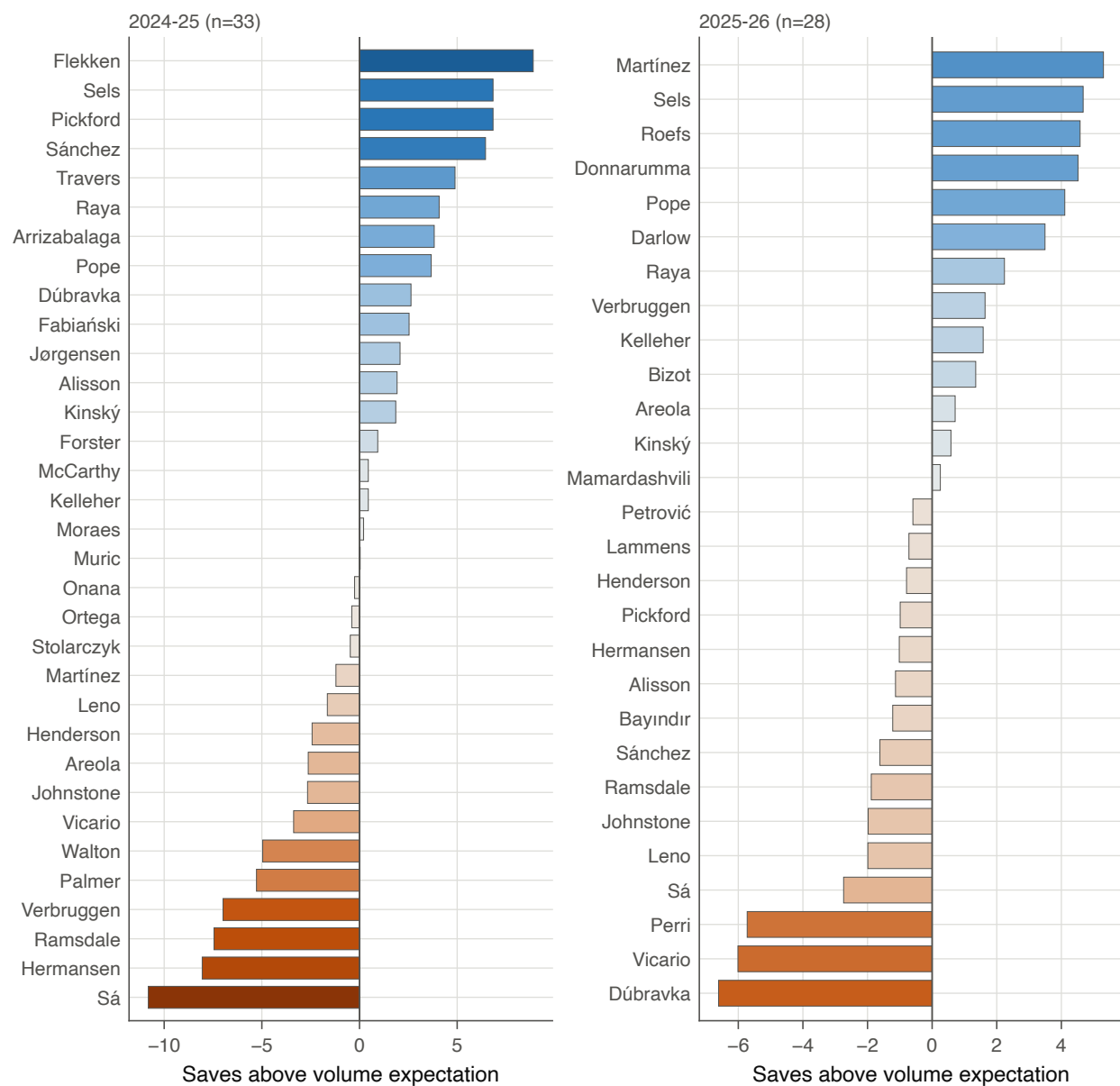


Figure S41: The shot stopping ranking that the two surviving reduced-sample composites support.

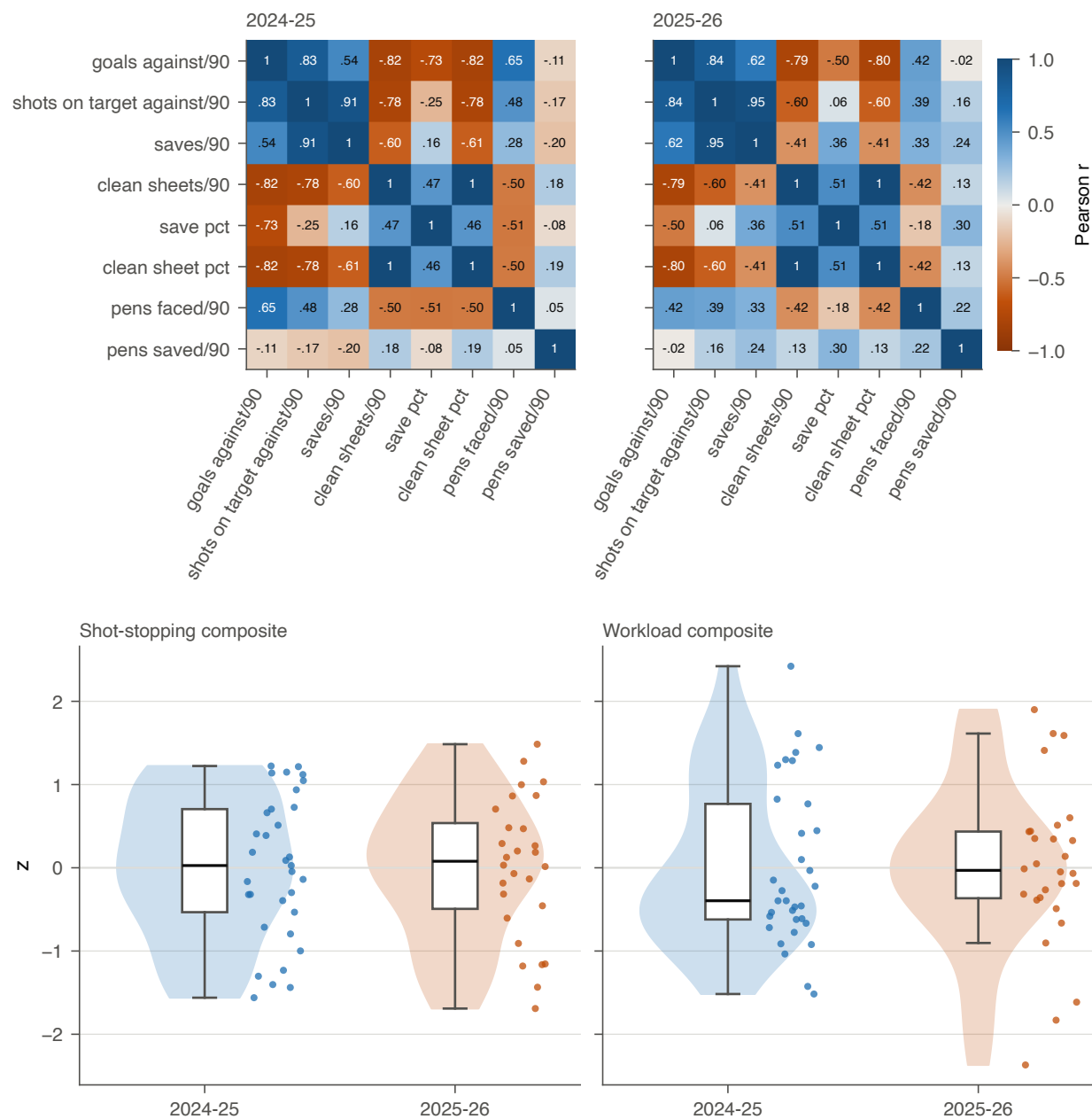


Figure S42: Reduced-sample goalkeeper features. Top, the correlation structure of what survived the withdrawal, which is what constrains the number of composites that can be built. Bottom, the distribution of the two composites.

S12 Cluster membership

Every player's archetype assignment is published as CSV in the accompanying repository rather than reproduced here, for both samples. The file `results/membership_primary.csv` carries the 9,263 eligible player-seasons of the primary sample with club, league, season, listed position, minutes and assigned archetype, and `results/membership_reduced.csv` carries the reduced sample with the same fields plus each player's distance from his cluster centroid. Both are regenerated by the pipeline, so they cannot drift from the numbers reported above.

References

- [1] Robert Tibshirani, Guenther Walther, and Trevor Hastie. Estimating the number of clusters in a data set via the gap statistic. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 63(2):411–423, 2001. doi: 10.1111/1467-9868.00293. URL <https://doi.org/10.1111/1467-9868.00293>.
- [2] Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2:193–218, 1985. doi: 10.1007/BF01908075. URL <https://doi.org/10.1007/BF01908075>.
- [3] Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou. Isolation forest. In *Proceedings of the 2008 Eighth IEEE International Conference on Data Mining (ICDM)*, pages 413–422. IEEE, 2008. doi: 10.1109/ICDM.2008.17. URL <https://doi.org/10.1109/ICDM.2008.17>.
- [4] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017. URL <https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html>.
- [5] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html. arXiv:1705.07874.
- [6] Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction, 2018. URL <https://arxiv.org/abs/1802.03426>.
- [7] Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9:2579–2605, 2008. URL <https://jmlr.org/papers/v9/vandermaaten08a.html>.
- [8] Bradley Efron and Carl Morris. Stein’s paradox in statistics. *Scientific American*, 236(5):119–127, 1977.
- [9] Ted Knutson. Introducing possession-adjusted player stats. StatsBomb, June 2014. URL <https://blogarchive.statsbomb.com/articles/soccer/introducing-possession-adjusted-player-stats/>. Published 23 June 2014.